Journal of Scientific Computing (2025) 105:98
https://doi.org/10.1007/s10915-025-03107-5

A high order Lagrangian type discontinuous Galerkin scheme for the compressible Euler equations on straight-edge meshes

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Received: 20 May 2025 / Revised: 27 September 2025 / Accepted: 19 October 2025 /
Published online: 22 November 2025
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Abstract

In this paper, we construct a high-order conservative cell-centered Lagrangian type discontinuous Galerkin (DG) scheme for the two-dimensional compressible Euler equations on straight-edge meshes. This high-order scheme is a highly non-trivial extension of the second-order pure Lagrangian DG scheme in our earlier work [15]. Unlike the previous second order pure Lagrangian methods, the basic idea of this high order scheme is that only nodes move in a pure Lagrangian fashion with the fluid velocity, while points connecting two nodes as a straight-line edge are subject to the flux exchange to maintain high order accuracy. On the cell edges containing the material interfaces and contact discontinuities, the flux term of this DG scheme is designed following that of our previous second-order pure Lagrangian scheme to strictly ensure that there is no mass exchange across these edges. In this way, the Lagrangian type DG scheme on the straight-edge meshes can not only achieve arbitrarily high order accuracy in smooth regions away from material interfaces and contact discontinuities, but also capture the material interfaces or contact discontinuities automatically and sharply. This high-order Lagrangian type DG scheme avoids curvilinear meshes, thus it is simple to implement and can save computational cost effectively. Another contribution of this work is that the oscillation-free (OF) procedure is introduced for the first time into the Lagrangian type method. In the Lagrangian framework, the shape of the meshes constantly changes and

Research of the second author Juan Cheng is supported in part by NSFC grant 12571432 and National Key R&D Program of China No. 2023YFA1009003. Research of the third author Chi-Wang Shu in part by NSF grant DMS-2309249.

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could be extremely non-uniform, which causes difficulties in controlling numerical oscillations for high order approximations. In this work, the OF procedure does not depend on any problem-related parameters, can effectively suppress oscillation near discontinuities without compromising the original high-order accuracy and conservation properties, and avoids additional time step size constraint caused by the numerical damping. Finally, a series of numerical experiments are presented to demonstrate the good performance of our high-order Lagrangian type DG scheme in terms of accuracy, resolution for discontinuities, and non-oscillation.

Keywords Lagrangian type DG scheme · Straight-edge mesh · High order accuracy · Conservation · Oscillation-free · Compressible Euler equations

1 Introduction

It is well known that there are two numerical frameworks for simulating fluid flow: the Eulerian framework and the Lagrangian framework. The Eulerian framework is based on the fixed grid, so it is easy to achieve arbitrarily high order accuracy. However, the Eulerian framework usually requires a very fine mesh to obtain high resolution near the material interfaces, resulting in excessive computational cost. In contrast, the Lagrangian framework allows the grid to move with the fluid velocity, which means that there is no mass exchange across the cell edges. Therefore, the Lagrangian framework has the significant advantage in tracking material interfaces automatically and sharply, and has become the most natural choice in the multi-material fluid flow fields such as computational fluid dynamics (CFD), inertial confinement fusion (ICF), and astrophysics.

Usually, the Lagrangian methods can be implemented in two ways, i.e., the staggered-grid method and the cell-centered method. The staggered-grid method [1, 3, 4, 24] refers to the kinematic variables (velocity) being stored at the vertices, the thermodynamic variables (density and pressure) being stored within the cells, whereas the cell-centered method [5, 32–34] refers to all variables being considered at the cell centers. In this work, we focus on the cell-centered Lagrangian method. Compared with the staggered-grid method, the cell-centered method does not require extra artificial viscosity, and it is easier to maintain the conservation property, achieve high-order accuracy, and design the remapping procedures when necessary.

The cell-centered Lagrangian method was first studied on the finite volume schemes, such as [5, 7, 33]. Recently, more research interests are focused on the Lagrangian discontinuous Galerkin (DG) schemes. The DG method was first proposed in [37]. Later, Cockburn et al. [8–12] successfully developed it into the Runge-Kutta (RK) DG methods for solving time-dependent nonlinear problems such as the Burgers' equation, Euler equations, and shallow water equations. At present, the DG method has become one of the most popular and attractive numerical methods for hyperbolic equations, due to its distinguished advantages of stencil compactness, high parallelization, flexibility on complex geometry and on h-p adaptation.

In the past years, there have been some studies devoted to developing the cell-centered Lagrangian DG methods. In [42, 43], Vilar et al. proposed the total Lagrangian DG scheme, which is built on a fixed reference configuration. The variables between the reference and the physical configurations are related by the deformation gradient tensor. The variables solved directly are often primitive variables such as specific volume, velocity, and specific total energy. Similar research on the total Lagrangian DG scheme can be found in [20, 25, 26, 30]. Another strategy for the cell-centered Lagrangian DG method is the updated Lagrangian DG

scheme, e.g., see [44, 45]. The updated Lagrangian DG scheme refers to the discretization on the moving physical configuration, so it is simpler and more intuitive. The DG space is defined on the time-dependent moving mesh, rather than on a fixed reference frame, so there is no need to study the time evolution of the deformation gradient tensor between different configurations. The conservative variables are solved directly, so the conservation of mass, momentum, and total energy is maintained effortlessly. However, we found that this kind of schemes leads to some flaws in accuracy, and the second-order accuracy for certain variables may be lost. To overcome this shortcoming, we proposed a new second-order updated Lagrangian DG scheme on straight-edge meshes in [15]. This new updated Lagrangian DG scheme is truly second-order accurate for all variables without any mass exchange between neighboring cells. We also developed a particular Lagrangian DG scheme in the cylindrical geometry, which is designed to be able to preserve both the one-dimensional spherical symmetry and conservation properties. In addition, as verified in [5, 6], it is impossible to obtain a pure Lagrangian scheme with higher than second-order accuracy on the straight-line edged grid. Therefore, Lagrangian schemes that attempt to extend to higher order accuracy have all been studied on curved meshes, such as [6, 13, 27, 28, 43]. But the computational cost on curved meshes is much larger than that on straight-line meshes. Its implementation is more complex. Also, it is more difficult to design mesh rezoning and physical variable remapping on curved meshes than on straight-line meshes. Therefore, it is challenging to construct a low-cost and low-complexity high-order Lagrangian type method.

Another crucial aspect of the Lagrangian DG method is to control the non-physical oscillations. Many numerical schemes may generate oscillations near the discontinuities, which challenges the robustness of the schemes. These numerical oscillations may cause non-physical structures and even lead to the explosion of numerical solutions. This is particularly severe for the Lagrangian method, as the mesh moves with the numerical solution of the fluid velocity.

In order to suppress the spurious oscillations, many techniques have been proposed in the literature. One approach is to apply slope limiters, including the minmod type total variation diminishing (TVD) limiter, total variation bounded (TVB) limiter, and weighted essentially nonoscillatory (WENO) limiter, see, e.g., [36, 39, 47]. Another alternative approach is to introduce artificial viscosity terms in the original schemes, such as [17, 18, 46]. More recently, in [29, 31], the authors proposed a different approach, called the oscillation-free (OF) DG method, in which a numerical damping term is cleverly added to the DG scheme. The added damping term can not only suppress the spurious oscillations but also preserve beneficial properties of the original DG scheme such as conservation, L^2-boundedness, optimal error estimates and superconvergence. For some strong discontinuous problems, the damping terms become quite stiff, resulting in very small time step sizes in traditional Runge-Kutta methods. To relax the time step size restrictions, the authors adopted the (modified) exponential Runge-Kutta method [19]. Motivated by [29, 31], the authors of [35] developed an oscillation-eliminating (OE) DG method, in which a new damping operator with scale-invariant and evolution-invariant properties is designed. By computing the original semi-discrete DG scheme and the damping equation in a split fashion, the OEDG scheme remains stable with the normal time step sizes. To the best knowledge of the authors, even though the OFDG and OEDG methods have been successfully applied to numerical simulations of many models, they have yet to be verified for the performance of controlling oscillations on moving non-uniform meshes, especially for the Lagrangian type DG methods.

In this paper, we will construct a new high-order conservative Lagrangian type discontinuous Galerkin scheme for the two-dimensional compressible Euler equations on straight-edge quadrilateral meshes. Our main contribution is twofold.

Firstly, the high-order Lagrangian type DG scheme contains an important ingredient, which is a highly non-trivial extension of that in [15]. In [15], we developed a second-order pure Lagrangian DG scheme. To ensure both second-order accuracy for all variables and no mass exchange between neighboring cells, we compute the $\mathbf{u} - \mathbf{u}_g$ flux term only to equations where the test functions are not equal to 1, where $\mathbf{u}$ is the fluid velocity and $\mathbf{u}_g$ is the grid velocity. By doing so, this Lagrangian DG scheme on straight-edge meshes is limited to second-order accuracy. In fact, the advantages of the pure Lagrangian methods are essentially concentrated at the material interfaces and contact discontinuities. With this in mind, we propose a different Lagrangian type framework, in which only nodes move exactly according to the fluid velocity (therefore ensuring that the method is an Lagrangian method), while the points connecting two nodes as a straight-line edge are subject to the $\mathbf{u}-\mathbf{u}_g$ flux term. On the cell edges containing the material interfaces or contact discontinuities, we throw away this flux term in the cell average equations to strictly ensure that there is no mass exchange across these edges. Since the grid nodes move with the fluid velocity, the cell edges containing the material interfaces or contact discontinuities are precisely identified by the logical number of the initial cell edges aligned with the initial discontinuities. In this new framework, although the high-order Lagrangian type DG scheme cannot be classified as a pure Lagrangian method, at the material interface and contact discontinuities, it automatically becomes a second-order pure Lagrangian DG method without mass exchange, the same as the second order version constructed in [15]. Therefore, this Lagrangian type DG scheme on the straight-edge meshes can not only achieve arbitrarily high order accuracy in smooth regions away from material interfaces and contact discontinuities, but also capture the material interfaces or contact discontinuities sharply. Compared with the high order Lagrangian scheme on curvilinear meshes, our designed high order scheme on straight-edge meshes can save computational cost effectively, is simple to implement, and allows easier design of rezoning and remapping steps when necessary. Note that our Lagrangian type DG scheme is fundamentally different from the arbitrary Lagrangian-Eulerian (ALE) DG schemes in [16, 22]. In our method, nodes move with the true fluid velocity to the designed high order accuracy, while the nodes in [16, 22] can only move in a straight line in space-time dictated by the fluid velocity at the current time step. Also, the DG schemes in [16, 22] did not take into account the special features at the material interfaces.

Another contribution of this paper is that the oscillation-free procedure is introduced for the first time into the Lagrangian type DG method to suppress numerical oscillations. In the Lagrangian type methods, the shape of the grid constantly changes and may be extremely non-uniform, which causes difficulties in controlling spurious oscillations. In this work, the damping term is designed by combining the OFDG scheme in [29] and the OEDG scheme in [35]. Firstly, the damping coefficients are given according to [29], where local mesh sizes between the neighboring cells and the characteristic decomposition are used to cater for the moving non-uniform grids. Then, the operator splitting strategy and exact solver [35] of the damping equation are used to avoid additional time step size constraint caused by numerical damping. This oscillation-free procedure can effectively control oscillation near the discontinuities without compromising the original high order accuracy and conservation properties. Also, it does not depend on any problem-related parameters, and is simple, efficient, and non-intrusive.

The rest of this paper is organized as follows. In Section 2, we review the second-order pure Lagrangian DG scheme, and then construct a new higher-order Lagrangian type DG

scheme on straight-edge meshes. In Section 3, we add the oscillation-free procedure to the DG method. A series of numerical experiments are presented in Section 4, to demonstrate the good performance of our high-order Lagrangian type DG scheme in terms of accuracy, resolution for discontinuities, and non-oscillation. Finally, we give concluding remarks in Section 5.

2 High order conservative Lagrangian type DG scheme for the two-dimensional Euler equations

2.1 The compressible Euler equations in two-dimensional Cartesian coordinates

The two-dimensional Euler equations for unsteady compressible flow can be described as

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}(\mathbf{U}) = 0, \ \ (x, y) \in \Omega(t), \ \ t \geq 0, \tag{2.1}$$

where $\Omega(t)$ is the moving spatial domain, $\mathbf{U}$ is the vector of the conserved variables, and $\mathbf{F}$ is the flux vector. Here $\mathbf{U}$ and $\mathbf{F}$ are defined as

$$\mathbf{U} = \begin{pmatrix} \rho \\ \rho \mathbf{u} \\ E \end{pmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{pmatrix} \rho \mathbf{u} \\ \rho \mathbf{u}\mathbf{u}^T + p\mathbf{I} \\ (E + p)\mathbf{u} \end{pmatrix}, \tag{2.2}$$

where ρ, $\mathbf{u} = (u, v)^T$, E, and p represent the density, velocity, total energy, and pressure, respectively. $\mathbf{I}$ is the 2×2 identity matrix. $\mathbf{M} = \rho \mathbf{u}$ is the momentum. The system is closed by adding an equation of state with the general form

$$p = p(\rho, e), \tag{2.3}$$

where $e = \frac{E}{\rho} - \frac{1}{2}|\mathbf{u}|^2$ is the specific internal energy. In particular, if we consider the perfect gas, then the equation of state is given by the following form

$$p = (\gamma - 1)\rho e = (\gamma - 1)(E - \frac{1}{2}\rho |\mathbf{u}|^2), \tag{2.4}$$

where γ is the specific gas constant.

This paper concentrates on the Lagrangian type framework, where the control volume moves with the fluid velocity. Therefore, the kinematic equation is written as

$$\frac{d\mathbf{x}}{dt} = \mathbf{u}, \quad \mathbf{x}(0) = \mathbf{x}_0, \tag{2.5}$$

where $\mathbf{x} = (x, y)^T$ and $\mathbf{x}_0$ denotes the coordinate at $t = 0$.

Next, for the above Euler equations, we will first review the second order pure Lagrangian DG scheme constructed in [15]. Then we will construct a new higher-order Lagrangian type DG scheme on the straight-edge grid. This higher-order DG scheme maintains the advantages of pure Lagrangian method at the material interfaces, can track the material interfaces automatically and sharply, and can maintain the optimal order accuracy in the smooth regions away from material interfaces and contact discontinuities.

2.2 The second-order conservative Lagrangian DG scheme

In this subsection, we give a brief review of the second-order conservative cell-centered pure Lagrangian DG scheme in [15], where the conserved variables $\mathbf{U}$ are solved directly. Let

us first introduce some notations used throughout the paper. We consider the quadrilateral meshes. The two-dimensional moving spatial domain $\Omega(t)$ is partitioned into $N \times M$ computational cells $I_{ij}(t)$ with $i = 1, \cdots, N,\ j = 1, \cdots, M$. The four vertices of the cell $I_{ij}(t)$ are denoted as $(x_{i-\frac{1}{2},j-\frac{1}{2}}, y_{i-\frac{1}{2},j-\frac{1}{2}})^T$, $(x_{i+\frac{1}{2},j-\frac{1}{2}}, y_{i+\frac{1}{2},j-\frac{1}{2}})^T$, $(x_{i+\frac{1}{2},j+\frac{1}{2}}, y_{i+\frac{1}{2},j+\frac{1}{2}})^T$ and $(x_{i-\frac{1}{2},j+\frac{1}{2}}, y_{i-\frac{1}{2},j+\frac{1}{2}})^T$. The edges of the cell I_{ij} are defined as ∂I_{ij}. We notice that the cell $I_{ij}(t)$ is time dependent. At the initial time, we use rectangular cells unless otherwise stated. The DG space is represented as

$$V_h^k(t) = \{\varphi(x, y, t) \in L^2(\Omega(t)) : \varphi(x, y, t)|_{I_{ij}(t)} \in \mathcal{P}^k(I_{ij}(t)),\ 1 \le i \le N,\ 1 \le j \le M\}, \tag{2.6}$$

where $\mathcal{P}^k(I_{ij}(t))$ denotes the set of polynomials of degree not greater than k in the cell I_{ij}, and the piecewise polynomial $\varphi \in V_h^k(t)$ is time-space dependent. Here and below, we use $\mathbf{U}$ to represent the numerical solutions for the conserved variables.

By using the Reynold's transport theorem and divergence theorem, we get the semi-discrete Lagrangian DG scheme for solving (2.1): find $\mathbf{U} \in \mathbf{V}_h^k(t)$ such that for all the test functions $\varphi(x, y, t) \in V_h^k(t)$ and $1 \le i \le N,\ 1 \le j \le M$, we have

$$\begin{aligned} \frac{d}{dt}\iint_{I_{ij}(t)} (\mathbf{U}\varphi(x, y, t))\, dxdy = &\iint_{I_{ij}(t)} \left(\mathbf{U}\left(\mathbf{u}\cdot\nabla\varphi + \frac{\partial\varphi}{\partial t}\right)\right) dxdy \\ &- \int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g)\cdot\mathbf{n}\varphi\right) ds \\ &+ \iint_{I_{ij}(t)} (\mathbf{G}\cdot\nabla\varphi)\, dxdy - \int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{G}}\cdot\mathbf{n}\varphi\right) ds, \end{aligned} \tag{2.7}$$

where $\mathbf{G}(\mathbf{U}) = (0, p\mathbf{I}, p\mathbf{u})^T$, $\mathbf{u}_g$ is the grid velocity, and $\mathbf{n} = (n_x, n_y)^T$ is the unit outward normal to ∂I_{ij}. $\widehat{\mathbf{G}}\cdot\mathbf{n} = (0, \widehat{p}\mathbf{n}, \widehat{p}\widehat{\mathbf{u}}\cdot\mathbf{n})^T$, $\widehat{\mathbf{U}} = (\widehat{\rho}, \widehat{\rho}\widehat{\mathbf{u}}, \widehat{E})^T$ and $\widehat{\mathbf{u}}\cdot\mathbf{n}$ are the numerical fluxes, which will be described in detail in the next subsection.

In the pure Lagrangian methods, the grid moves with the fluid velocity, and there should be no mass exchange across the moving cell edges. Therefore, we need to deal carefully with the flux term $\int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g)\cdot\mathbf{n}\varphi\right) ds$. If this flux term is thrown away directly, the accuracy of the DG scheme is only first-order for some variables, such as the conserved variables, even if we use high-order polynomials. According to [15], we assume that $\int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g)\cdot\mathbf{n}\right) ds = 0$ only in the cell average equations, i.e., this flux term $\int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g)\cdot\mathbf{n}\varphi\right) ds$ is only computed in the equations where the test functions are not equal to 1. In this way, there is no mass exchange between neighboring cells. The local mass conservation is valid. Since we cannot get higher than second order accuracy for the pure Lagrangian methods on the straight-line grid [5, 6], we set $k \le 1$ in this subsection. More details about this scheme can be found in [15].

2.3 The high order conservative Lagrangian type DG scheme

In the following, we will propose a new higher-order conservative cell-centered Lagrangian type DG scheme on the straight-line quadrilateral grid. We mainly focus on the case of $k > 1$. Differently from the subsection 2.2, we consider another Lagrangian type framework, in which only nodes move with the fluid velocity (thus justifying the word "Lagrangian"), however other points connecting two nodes as a straight-line edge are subject to the flux term $\int \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g)\cdot\mathbf{n}\varphi\right) ds$. The DG schemes under this framework can be arbitrarily high order

accurate for smooth solutions on the straight-edge grid. This is because the DG schemes thus constructed are the bona fide DG scheme in a moving mesh, consistent with the RKDG methods proposed by Cockburn et al. in [8–12], and hence naturally achieve high order accuracy.

At the contact discontinuities or material interfaces of multi-material flows, which can be trivially identified in the Lagrangian framework as their indices do not change with time, we adopt the strategy in the above second order pure Lagrangian DG scheme. First, we set the initial cell edges to be aligned with the initial discontinuities, i.e., there are no mixed cells. We label the set of these cell edges as Γ_c. Then the cell edges in the set Γ_c are always located at the contact discontinuities or material interfaces as time evolves. We take the DG basis functions to be 1 and those orthogonal to 1, and on Γ_c, we throw away the flux term $\int_{\Gamma_c} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g) \cdot \mathbf{n}\right) ds$ in the cell average equations (i.e. those with the test function $\varphi = 1$), to ensure that there is no mass exchange across these edges. That is, the flux term $\int_{\Gamma_c} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g) \cdot \mathbf{n}\varphi\right) ds$ is only computed in the equations where the basis functions are not equal to 1.

While it is not necessary to remove the flux term $\int \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g) \cdot \mathbf{n}\right) ds$ to the cell average equations at shocks within a single material (as mass exchange at such shocks within a single material does not lead to severe consequences), we still remove it at such shocks since we find that removing this flux term at shocks can help enhance the robustness of the grid movement in numerical simulations. For such purpose, we first need to determine the cell edges containing the shocks. In this paper, we apply the KXRCF detection technique [23] to identify these edges. Let l be a cell edge. Define u_h^- and u_h^+ as the upwind ($(\widehat{\mathbf{u}} - \mathbf{u}_g) \cdot \mathbf{n} > 0$) and downwind ($(\widehat{\mathbf{u}} - \mathbf{u}_g) \cdot \mathbf{n} < 0$) sides of the indicator variable u_h along l, respectively. The cell edge l is marked in the set Γ_s when

$$\frac{|\int_l (u_h^- - u_h^+) ds|}{|l|^{\frac{k+3}{2}} \|u_h^-\|_{L^\infty(l)}} > C_k, \tag{2.8}$$

where C_k is a constant, usually, we take $C_k = 1$. $\|\cdot\|_{L^\infty(l)}$ is the standard L^∞ norm on the edge l. Here we take the pressure p as the indicator variable. When $l \in \Gamma_s$, we ignore this flux term $\int_l \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g) \cdot \mathbf{n}\right) ds$ in the cell average equations, similar to the treatment at the material interfaces.

With the above adjustments, our semi-discrete high-order Lagrangian type DG scheme can be written as follows: find $\mathbf{U} \in \mathbf{V}_h^k(t)$ to satisfy the following semi-discrete scheme

$$\frac{d}{dt} \iint_{I_{ij}(t)} \mathbf{U} dx dy = -\int_{\partial I_{ij}(t) \setminus \Gamma_{cs}^{ij}} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g) \cdot \mathbf{n}\right) ds - \int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{G}} \cdot \mathbf{n}\right) ds \tag{2.9}$$

$$\begin{aligned} \frac{d}{dt} \iint_{I_{ij}(t)} (\mathbf{U}\phi_m)\, dx dy = & \iint_{I_{ij}(t)} \left(\mathbf{U}\left(\mathbf{u} \cdot \nabla \phi_m + \frac{\partial \phi_m}{\partial t}\right)\right) dx dy \\ & - \int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g) \cdot \mathbf{n}\phi_m\right) ds \\ & + \iint_{I_{ij}(t)} (\mathbf{G} \cdot \nabla \phi_m)\, dx dy - \int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{G}} \cdot \mathbf{n}\phi_m\right) ds, \end{aligned} \tag{2.10}$$

where $m = 1, \cdots, K$ and $K = \frac{(k+1)(k+2)}{2} - 1$. Let $\phi_0 = 1$, $\{\phi_0, \phi_1, \cdots, \phi_K\}$ are the $K+1$ basis functions in the DG space V_h^k, and $\{\phi_1, \cdots, \phi_K\}$ are orthogonal to $\phi_0 = 1$. We take $\Gamma_{cs}^{ij} = \partial I_{ij} \cap \{\Gamma_c \cup \Gamma_s\}$. Γ_c is the set of cell edges containing the contact discontinuities and material interfaces, and Γ_s is the set of cell edges containing shock waves.

Remark 2.1 The above high order DG scheme (2.9)-(2.10) is not a pure Lagrangian method because there is mass exchange between the neighboring cells in the smooth regions (edges not belonging to Γ_{cs}^{ij}). However, the nodes are always moving according to the fluid velocity up to the designed high order accuracy, thus justifying the word "Lagrangian". Also, at the contact discontinuities and material interfaces, the above scheme automatically becomes a pure Lagrangian scheme, thus they can be captured sharply.

Next, let us turn to other details in this high order DG scheme, including the numerical fluxes, the time derivatives of basis functions, and the grid velocity.

Define $(\rho^{int}, \mathbf{u}^{int}, p^{int})^T$ and $(\rho^{out}, \mathbf{u}^{out}, p^{out})^T$ as the variables inside and outside the cell I_{ij} (inside the neighboring cell) along the cell edges ∂I_{ij}, respectively. We need to determine the numerical fluxes $\widehat{\mathbf{G}} \cdot \mathbf{n} = (0, \widehat{p}\mathbf{n}, \widehat{p}\widehat{\mathbf{u}} \cdot \mathbf{n})^T$ and $\widehat{\mathbf{U}} = (\widehat{\rho}, \widehat{\rho}\widehat{\mathbf{u}}, \widehat{E})^T$ to ensure stability and accuracy. Similar to that in [15], we use the Harten-Lax-van Leer Contact (HLLC) approximate Riemann solver [7] in the Lagrangian description for $\widehat{\mathbf{G}} \cdot \mathbf{n}$. The HLLC numerical flux has a very simple form in the Lagrangian framework. We take $u_n^{int/out} = \mathbf{u}^{int/out} \cdot \mathbf{n}$ and $c^{int/out} = \sqrt{\frac{\gamma p^{int/out}}{\rho^{int/out}}}$. We can obtain μ_* and p_* as follows:

$$\begin{aligned} \mu_* &= \frac{\rho^{out} u_n^{out}(\mu_+ - u_n^{out}) - \rho^{int} u_n^{int}(\mu_- - u_n^{int}) + p^{int} - p^{out}}{\rho^{out}(\mu_+ - u_n^{out}) - \rho^{int}(\mu_- - u_n^{int})}, \\ p_* &= \rho^{int}(u_n^{int} - \mu_-)(u_n^{int} - \mu_*) + p^{int}, \end{aligned} \tag{2.11}$$

where μ_- and μ_+ represent the acoustic wavespeeds with

$$\mu_- = \min\{u_n^{int} - c^{int},\ \tilde{u}_n - \tilde{c}\},\ \ \mu_+ = \max\{u_n^{out} + c^{out},\ \tilde{u}_n + \tilde{c}\},$$

and $\tilde{u}_n$ and $\tilde{c}$ are the Roe averages of u_n and c respectively. Thus we have

$$\widehat{\mathbf{G}} \cdot \mathbf{n} = \left(0,\ p_*\mathbf{n},\ p_*\mu_*\right)^T. \tag{2.12}$$

For the other flux $\widehat{\mathbf{U}} = (\widehat{\rho}, \widehat{\rho}\widehat{\mathbf{u}}, \widehat{E})^T$, we choose the upwind flux, namely,

$$(\widehat{\rho}, \widehat{\rho}\widehat{\mathbf{u}}, \widehat{E}) = \begin{cases} (\rho^{int}, \rho^{int}\mathbf{u}^{int}, E^{int}), & \text{if } \mu_* - \mathbf{u}_g \cdot \mathbf{n} \geq 0, \\ (\rho^{out}, \rho^{out}\mathbf{u}^{out}, E^{out}), & \text{if } \mu_* - \mathbf{u}_g \cdot \mathbf{n} < 0. \end{cases} \tag{2.13}$$

For the time derivatives of basis functions $\frac{\partial \phi_m}{\partial t}$ in the scheme (2.10), we get it by a similar derivation as in [15]. Taking $k = 2$ as an example, without loss of generality, we assume that the basis functions of the V_h^2 space in the cell I_{ij} are as follows

$$\begin{aligned} \phi_0^{ij} &= 1, \\ \phi_1^{ij}(x, y, t) &= 2\frac{x - \overline{x}_{1c}(t)}{h_x(t)}, \\ \phi_2^{ij}(x, y, t) &= 2\frac{y - \overline{y}_{1c}(t)}{h_y(t)}, \\ \phi_3^{ij}(x, y, t) &= 2\frac{(x - \overline{x}_{1c}(t))^2 - \overline{x}_{2c}(t) + \overline{x}_{1c}^2(t)}{h_x^2(t)}, \\ \phi_4^{ij}(x, y, t) &= 2\frac{(y - \overline{y}_{1c}(t))^2 - \overline{y}_{2c}(t) + \overline{y}_{1c}^2(t)}{h_y^2(t)}, \\ \phi_5^{ij}(x, y, t) &= 4\frac{(x - \overline{x}_{1c}(t))(y - \overline{y}_{1c}(t)) - \overline{xy}_{2c}(t) + \overline{x}_{1c}(t)\overline{y}_{1c}(t)}{h_x(t)h_y(t)}, \end{aligned} \tag{2.14}$$

where $h_x(t) = x_{i+\frac{1}{2},j+\frac{1}{2}}(t) - x_{i-\frac{1}{2},j-\frac{1}{2}}(t)$, $h_y(t) = y_{i+\frac{1}{2},j+\frac{1}{2}}(t) - y_{i-\frac{1}{2},j-\frac{1}{2}}(t)$, $\overline{x}_{1c} = \frac{\iint_{I_{ij}(t)} x dxdy}{S_{ij}(t)}$, $\overline{y}_{1c} = \frac{\iint_{I_{ij}(t)} y dxdy}{S_{ij}(t)}$, $\overline{x}_{2c} = \frac{\iint_{I_{ij}(t)} x^2 dxdy}{S_{ij}(t)}$, $\overline{y}_{2c} = \frac{\iint_{I_{ij}(t)} y^2 dxdy}{S_{ij}(t)}$, $\overline{xy}_{2c} = \frac{\iint_{I_{ij}(t)} xy dxdy}{S_{ij}(t)}$, and S_{ij} represents the area of the cell $I_{ij}(t)$. We notice that $\{\phi_m^{ij}(x, y, t),\ m = 1, \cdots, 5\}$ are orthogonal to $\phi_0^{ij} = 1$. For simplicity, we omit the superscript 'ij' and the time variable t in the following formulas. Using the Reynold's transport theorem, we can get

$$
\begin{aligned}
\overline{u}_{1c} &\equiv \frac{d\overline{x}_{1c}}{dt} = \frac{1}{S}\int_{\partial I(t)} (x - \overline{x}_{1c})\mathbf{u}_g \cdot \mathbf{n} ds, \\
\overline{v}_{1c} &\equiv \frac{d\overline{y}_{1c}}{dt} = \frac{1}{S}\int_{\partial I(t)} (y - \overline{y}_{1c})\mathbf{u}_g \cdot \mathbf{n} ds, \\
\overline{u}_{2c} &\equiv \frac{d\overline{x}_{2c}}{dt} = \frac{1}{S}\int_{\partial I(t)} (x^2 - \overline{x}_{2c})\mathbf{u}_g \cdot \mathbf{n} ds, \\
\overline{v}_{2c} &\equiv \frac{d\overline{y}_{2c}}{dt} = \frac{1}{S}\int_{\partial I(t)} (y^2 - \overline{y}_{2c})\mathbf{u}_g \cdot \mathbf{n} ds, \\
\overline{uv}_{2c} &\equiv \frac{d\overline{xy}_{2c}}{dt} = \frac{1}{S}\int_{\partial I(t)} (xy - \overline{xy}_{2c})\mathbf{u}_g \cdot \mathbf{n} ds,
\end{aligned}
\tag{2.15}
$$

where $\mathbf{u}_g \cdot \mathbf{n}$ represents the grid velocity along the outward normal direction of the cell edges ∂I, which is explained in detail in the next subsection. Using (2.15), $\frac{\partial \phi_m}{\partial t}$ is easily obtained through the following formulas

$$
\begin{aligned}
\frac{\partial \phi_0}{\partial t} &= 0, \\
\frac{\partial \phi_1}{\partial t} &= -2\left(\frac{\overline{u}_{1c}}{h_x} + \frac{(x - \overline{x}_{1c})\Delta u}{h_x^2}\right), \\
\frac{\partial \phi_2}{\partial t} &= -2\left(\frac{\overline{v}_{1c}}{h_y} + \frac{(y - \overline{y}_{1c})\Delta v}{h_y^2}\right), \\
\frac{\partial \phi_3}{\partial t} &= -2\frac{2(x - 2\overline{x}_{1c})\overline{u}_{1c} + \overline{u}_{2c}}{h_x^2} - 4\frac{((x - \overline{x}_{1c})^2 - \overline{x}_{2c} + \overline{x}_{1c}^2)\Delta u}{h_x^3}, \\
\frac{\partial \phi_4}{\partial t} &= -2\frac{2(y - 2\overline{y}_{1c})\overline{v}_{1c} + \overline{v}_{2c}}{h_y^2} - 4\frac{((y - \overline{y}_{1c})^2 - \overline{y}_{2c} + \overline{y}_{1c}^2)\Delta v}{h_y^3}, \\
\frac{\partial \phi_5}{\partial t} &= -4\frac{(x - 2\overline{x}_{1c})\overline{v}_{1c} + (y - 2\overline{y}_{1c})\overline{u}_{1c} + \overline{uv}_{2c}}{h_x h_y} \\
&\quad - 4\left((x - \overline{x}_{1c})(y - \overline{y}_{1c}) - \overline{xy}_{2c} + \overline{x}_{1c}\overline{y}_{1c}\right)\left(\frac{\Delta v}{h_x h_y^2} + \frac{\Delta u}{h_x^2 h_y}\right),
\end{aligned}
\tag{2.16}
$$

where $\Delta u = u_{i+\frac{1}{2},j+\frac{1}{2}} - u_{i-\frac{1}{2},j-\frac{1}{2}}$, $\Delta v = v_{i+\frac{1}{2},j+\frac{1}{2}} - v_{i-\frac{1}{2},j-\frac{1}{2}}$. $u_{i\pm\frac{1}{2},j\pm\frac{1}{2}}$ and $v_{i\pm\frac{1}{2},j\pm\frac{1}{2}}$ are the vertex velocities in the x and y directions, respectively, which are defined in detail in the next subsection.

2.4 The determination of the grid velocity

Now, we first introduce how to determine the vertex velocity $\mathbf{u}_{i-\frac{1}{2},j-\frac{1}{2}} = (u_{i-\frac{1}{2},j-\frac{1}{2}}, v_{i-\frac{1}{2},j-\frac{1}{2}})^T$, which is used in the node motion equation (2.5) and the time derivatives of the basis functions (2.16). Then we derive an expression of the grid velocity $\mathbf{u}_g$ on the cell edges.

In this paper, we consider the vertex velocity designed in [5]. Briefly, we consider a vertex $(x_{i-\frac{1}{2},j-\frac{1}{2}}, y_{i-\frac{1}{2},j-\frac{1}{2}})^T$ shared by four edges with a serial number $l = 1, 2, 3, 4$. The normal velocity of this vertex along each edge l is defined as $u_n^l = \widehat{\mathbf{u}} \cdot \mathbf{n}_l = \mu_*$, where $\mathbf{n}_l = (n_x^l, n_y^l)^T$ is the unit normal of the edge l. For the tangential velocity, we have

$$u_t^l = \frac{1}{2}(u_t^{l-} + u_t^{l+}),\ l = 1, 2, 3, 4,$$

where $u_t^{l\pm} = \mathbf{u}^{l\pm} \cdot \mathbf{t}_l$, $\mathbf{u}^{l\pm}$ is the left/right velocity at this vertex along the edge l, and $\mathbf{t}_l = (t_x^l, t_y^l)^T$ is the unit tangential of the edge l. Then we transform u_t^l and u_n^l into the x-y velocity components, which have the following formulas,

$$u_x^l = \frac{u_n^l t_y^l - u_t^l n_y^l}{n_x^l t_y^l - n_y^l t_x^l},\quad u_y^l = \frac{-u_n^l t_x^l + u_t^l n_x^l}{n_x^l t_y^l - n_y^l t_x^l},\ l = 1, 2, 3, 4.$$

Finally, we can get the vertex velocity as follows,

$$u_{i-\frac{1}{2},j-\frac{1}{2}} = \frac{1}{4}\sum_{l=1}^{4} u_x^l,\quad v_{i-\frac{1}{2},j-\frac{1}{2}} = \frac{1}{4}\sum_{l=1}^{4} u_y^l. \tag{2.17}$$

Using the vertex velocity (2.17), we can derive an expression of the grid velocity $\mathbf{u}_g = (u_g, v_g)^T$ on the cell edges, which is used in the Lagrangian type DG scheme (2.9)-(2.10). In this work, we consider the straight-edge mesh. In order to ensure that the grids have straight-line edges during the movement, we can obtain the following expression of the grid velocity on the edge l,

$$u_g = u_1\frac{x - x_2}{x_1 - x_2} + u_2\frac{x - x_1}{x_2 - x_1},\quad v_g = v_1\frac{y - y_2}{y_1 - y_2} + v_2\frac{y - y_1}{y_2 - y_1}, \tag{2.18}$$

where the point $(x, y)^T$ is located on the cell edge l from $(x_1, y_1)^T$ to $(x_2, y_2)^T$. $(u_1, v_1)^T$ and $(u_2, v_2)^T$ are the vertex velocities at the two endpoints of the cell edge l, which are calculated in the formula (2.17). It should be noted that the mesh movement velocity on the cell edge is calculated through a linear combination of the vertex velocities at the two endpoints in (2.18). This does not affect the high-order accuracy of the scheme, since the difference between the fluid velocity and mesh movement velocity along the straightline cell edge has already been taken into account in the scheme.

Up to now, we have described all the details in the semi-discrete high-order Lagrangian type DG scheme. Next, we introduce the high order time discretization method in order to obtain a fully discrete numerical scheme.

2.5 Time discretization

In the Lagrangian simulation, we apply the third-order strong stability preserving (SSP) Runge-Kutta method [40] to the time marching for the semi-discrete DG scheme (2.9)-(2.10)

and the node motion equation (2.5). Specifically, the third-order SSP-RK time discretization has the following three stages.

Stage 1:

$$\begin{aligned} \mathbf{x}^{n,1}_{i-\frac{1}{2},j-\frac{1}{2}} &= \mathbf{x}^{n}_{i-\frac{1}{2},j-\frac{1}{2}} + \Delta t^n \mathbf{u}^{n}_{i-\frac{1}{2},j-\frac{1}{2}}, \\ \mathbf{A}^{n,1}\mathbf{U}^{n,1}_{c} &= \mathbf{A}^{n}\mathbf{U}^{n}_{c} + \Delta t^n \mathbf{L}(\mathbf{U}^{n}_{c}); \end{aligned} \tag{2.19}$$

Stage 2:

$$\begin{aligned} \mathbf{x}^{n,2}_{i-\frac{1}{2},j-\frac{1}{2}} &= \frac{3}{4}\mathbf{x}^{n}_{i-\frac{1}{2},j-\frac{1}{2}} + \frac{1}{4}\mathbf{x}^{n,1}_{i-\frac{1}{2},j-\frac{1}{2}} + \frac{1}{4}\Delta t^n \mathbf{u}^{n,1}_{i-\frac{1}{2},j-\frac{1}{2}}, \\ \mathbf{A}^{n,2}\mathbf{U}^{n,2}_{c} &= \frac{3}{4}\mathbf{A}^{n}\mathbf{U}^{n}_{c} + \frac{1}{4}\mathbf{A}^{n,1}\mathbf{U}^{n,1}_{c} + \frac{1}{4}\Delta t^n \mathbf{L}(\mathbf{U}^{n,1}_{c}); \end{aligned} \tag{2.20}$$

Stage 3:

$$\begin{aligned} \mathbf{x}^{n+1}_{i-\frac{1}{2},j-\frac{1}{2}} &= \frac{1}{3}\mathbf{x}^{n}_{i-\frac{1}{2},j-\frac{1}{2}} + \frac{2}{3}\mathbf{x}^{n,2}_{i-\frac{1}{2},j-\frac{1}{2}} + \frac{2}{3}\Delta t^n \mathbf{u}^{n,2}_{i-\frac{1}{2},j-\frac{1}{2}}, \\ \mathbf{A}^{n+1}\mathbf{U}^{n+1}_{c} &= \frac{1}{3}\mathbf{A}^{n}\mathbf{U}^{n}_{c} + \frac{2}{3}\mathbf{A}^{n,2}\mathbf{U}^{n,2}_{c} + \frac{2}{3}\Delta t^n \mathbf{L}(\mathbf{U}^{n,2}_{c}); \end{aligned} \tag{2.21}$$

where the spatial operator $\mathbf{L}$ represents the right hand terms of the scheme (2.9)-(2.10), $\mathbf{A}$ is the mass matrix, and $\mathbf{U}_c$ represents the coefficients of the numerical solution polynomials for any conserved variables in the cell I_{ij}. Here the variables with the superscripts n and $n+1$ represent the corresponding values at the n-th and $(n+1)$-th time steps, respectively, and we omit the subscript 'ij' in $\mathbf{A}$ and $\mathbf{U}_c$. Notice that the vertex velocity, node position and numerical solution polynomials are updated at each RK stage in the above time method (2.19)-(2.21). The time step size Δt^n is determined as

$$\Delta t^n = \lambda \min_{i,j} \frac{\Delta l_{ij}}{c^n_{ij} + |\mathbf{u}|^n_{ij}}, \tag{2.22}$$

where Δl_{ij} is the shortest edge length of the cell $I_{ij}(t^n)$ and λ is the Courant number.

So far, we have a fully discrete $(k+1)$-order Lagrangian type scheme.

3 High order conservative Lagrangian type oscillation-free DG scheme

The DG scheme (2.9)-(2.10) may generate spurious oscillations near the discontinuities, which makes the scheme less robust and may even lead to a crash of the code. To overcome this difficulty, we follow the idea of the OFDG method in [29, 31]. This OFDG method does not need any parameters related to the problem, can automatically assess the intensity of the discontinuity and suppress numerical oscillations near the strong discontinuities while maintaining the original high order accuracy for smooth solutions.

3.1 The semi-discrete Lagrangian type OFDG scheme

In this subsection, we will construct a semi-discrete Lagrangian type OFDG scheme, which essentially adds an extra damping term to control the spurious oscillations based on the original high-order Lagrangian type DG scheme (2.9)-(2.10).

The semi-discrete OFDG scheme for solving (2.1) is as follows: seek $\mathbf{U} \in \mathbf{V}_h^k(t)$ such that for $\forall \varphi(x, y, t) \in V_h^k(t)$,

$$\begin{aligned}
&\frac{d}{dt}\iint_{I_{ij}(t)} (\mathbf{U}\varphi(x, y, t))\, dxdy \\
&= \iint_{I_{ij}(t)} \left(\mathbf{U}\left(\mathbf{u}\cdot\nabla\varphi + \frac{\partial\varphi}{\partial t}\right)\right) dxdy - \int_{\partial I_{ij}^{Ncs}(t)} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g)\cdot\mathbf{n}\varphi\right) ds \\
&\quad + \iint_{I_{ij}(t)} (\mathbf{G}\cdot\nabla\varphi)\, dxdy - \int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{G}}\cdot\mathbf{n}\varphi\right) ds \\
&\quad - \sum_{a=0}^{k} \frac{\sigma_{ij}^a}{h_{ij}} \iint_{I_{ij}(t)} \left(\mathbf{U} - P_h^{a-1}\mathbf{U}\right)\varphi dxdy,
\end{aligned} \tag{3.1}$$

where $\partial I_{ij}^{Ncs} = \partial I_{ij}\backslash\Gamma_{cs}^{ij}$ when $\varphi = 1$, otherwise $\partial I_{ij}^{Ncs} = \partial I_{ij}$, see the DG scheme (2.9)-(2.10) for details. Here we call $-\sum_{a=0}^{k} \frac{\sigma_{ij}^a}{h_{ij}} \iint_{I_{ij}(t)} \left(\mathbf{U} - P_h^{a-1}\mathbf{U}\right)\varphi dxdy$ the damping term. h_{ij} is the longest edge length of the cell I_{ij}. P_h^a is the standard L^2 projection into $\mathbf{V}_h^a$, $a \geq 0$, i.e., for any vector function $\mathbf{w}$, $P_h^a\mathbf{w} \in \mathbf{V}_h^a$ satisfies

$$\iint_{I_{ij}} (P_h^a\mathbf{w} - \mathbf{w})\varphi dxdy = 0,\ \forall\varphi \in p^a(I_{ij}), \tag{3.2}$$

and we set $P_h^{-1} = P_h^0$. By (3.2), we can find that $\iint_{I_{ij}(t)} \left(\mathbf{U} - P_h^{a-1}\mathbf{U}\right) dxdy = 0,\ a = 0, 1, \cdots, k$, that is, the damping term does not play a role in the cell average equations, so the conservation of the above scheme is not lost.

One key aspect of controlling numerical oscillations is the design of the damping coefficients σ_{ij}^a. These coefficients σ_{ij}^a are non-negative, small in the smooth regions and large near the discontinuities. Referring to [29], the damping coefficients are given as follows:

$$\sigma_{ij}^a = \frac{2(2a+1)}{(2k-1)} \frac{\widetilde{h}_{ij}^a}{a!} \max_{1\leq q\leq 4} \sum_{|\boldsymbol{\alpha}|=a} \left(\frac{1}{N_e} \sum_{\boldsymbol{v}\in\partial I_{ij}} \left(\left[\!\left[\partial^{\boldsymbol{\alpha}} V_q\right]\!\right]\Big|_{\boldsymbol{v}}\right)^2\right)^{\frac{1}{2}}, \tag{3.3}$$

where $\widetilde{h}_{ij} = \max\{h_{ij}, h_{i\pm1,j}, h_{i,j\pm1}\}$. The vector $\boldsymbol{\alpha} = (\alpha_1, \alpha_2)^T$ is the multi-index with $|\boldsymbol{\alpha}| = \alpha_1 + \alpha_2$ in two dimensions, and $\partial^{\boldsymbol{\alpha}} w$ is defined as

$$\partial^{\boldsymbol{\alpha}} w = \frac{\partial^{|\boldsymbol{\alpha}|} w}{\partial x^{\alpha_1}\partial y^{\alpha_2}}.$$

$\boldsymbol{v} = (x_{i\pm\frac{1}{2},j\pm\frac{1}{2}}, y_{i\pm\frac{1}{2},j\pm\frac{1}{2}})^T$ is the vertex of the cell I_{ij}. $\left[\!\left[w\right]\!\right]\Big|_{\boldsymbol{v}}$ represents the jump of the function w at the vertex $\boldsymbol{v}$ of two neighboring cells that share the same edge. N_e is the number of edges of the cell I_{ij}, and we take $N_e = 4$ in the quadrilateral mesh.

In this work, we use the local characteristic decomposition for the jumps of the damping coefficients. Define $\mathbf{R}^{-1}$ as a 4×4 matrix with the left eigenvectors of Jacobian matrix $\mathbf{F}(\overline{\mathbf{U}})\cdot\mathbf{n}_l$, where $\mathbf{n}_l = (n_x, n_y)^T,\ l = 1, 2, 3, 4$ is the outward unit normal to different edges of the cell I_{ij}, and $\overline{\mathbf{U}}$ is the cell average vector of the conserved variables. In the two-dimensional compressible Euler equations of the perfect gas, $\mathbf{R}^{-1}$ has the following

expression

$$\mathbf{R}^{-1}=\frac{\gamma-1}{c}\begin{pmatrix}\frac{1}{2}\left(B_r+\tilde{u}c\right) & -\frac{1}{2}\left((\gamma-1)u+n_xc\right) & -\frac{1}{2}\left((\gamma-1)v+n_yc\right) & \frac{1}{2}(\gamma-1)\\ \tilde{v}c & n_yc & -n_xc & 0\\ c^2-B_r & (\gamma-1)u & (\gamma-1)v & -(\gamma-1)\\ \frac{1}{2}\left(B_r-\tilde{u}c\right) & -\frac{1}{2}\left((\gamma-1)u-n_xc\right) & -\frac{1}{2}\left((\gamma-1)v-n_yc\right) & \frac{1}{2}(\gamma-1)\end{pmatrix}, \tag{3.4}$$

where $\tilde{u}=un_x+vn_y$, $\tilde{v}=-un_y+vn_x$, and $B_r=(\gamma-1)\frac{u^2+v^2}{2}$. Then the vector $\partial^{\boldsymbol{\alpha}}\boldsymbol{V}|_{\boldsymbol{v}}=(\partial^{\boldsymbol{\alpha}}V_1,\ldots,\partial^{\boldsymbol{\alpha}}V_4)^T\Big|_{\boldsymbol{v}}$ is obtained by $\partial^{\boldsymbol{\alpha}}\boldsymbol{V}|_{\boldsymbol{v}}=\mathbf{R}^{-1}\big|_{\mathbf{n}_l}\,\partial^{\boldsymbol{\alpha}}\mathbf{U}|_{\boldsymbol{v}}$.

Remark 3.1 In this paper, we use the local characteristic decomposition to get the damping coefficients. If we do not use the characteristic decomposition, we find that the numerical results may be much less satisfactory in some numerical examples, see, e.g. Fig. 9 of the blast waves problem.

Remark 3.2 It should be pointed out that $\widetilde{h}_{ij}$ in (3.3) is different from that in [29]. Here $\widetilde{h}_{ij}$ is the local mesh size between the neighboring cells rather than the global one. In the Lagrangian framework, the mesh shape constantly changes and may even be extremely non-uniform, so the choice of mesh size affects the performance of controlling numerical oscillations. We find that numerical oscillations could be effectively controlled without excessive smearing when the local mesh sizes are used in the numerical experiments.

3.2 The operator splitting strategy

In some strong discontinuity problems, the damping term may be very large, resulting in very small time step size in traditional RK time discretization. To relax time step size restriction, the authors in [29] adopted the (modified) exponential Runge-Kutta method. In this paper, we use the approach in [35], namely we first apply the operator splitting strategy to the above OFDG scheme (3.1), and then develop an exact solver for the damping equation. Compared with the exponential RK time discretization, this approach is stable with normal time step sizes, and the oscillation-free procedure can be embedded in the original DG algorithm as an independent module.

To be more specific, the semi-discrete Lagrangian type OFDG scheme (3.1) can be split into the original DG scheme (2.9)-(2.10) and the following damping equation. Find $\mathbf{U}_\sigma\in\mathbf{V}_h^k$, such that for $\forall\varphi\in V_h^k$,

$$\begin{cases}\dfrac{d}{d\tilde{t}}\displaystyle\iint_{I_{ij}^{n,s+1}}(\mathbf{U}_\sigma\varphi)\,dxdy+\sum_{a=0}^{k}\frac{\sigma_{ij}^a}{h_{ij}}\iint_{I_{ij}^{n,s+1}}\left(\mathbf{U}_\sigma-P_h^{a-1}\mathbf{U}_\sigma\right)\varphi dxdy=0,\\ \mathbf{U}_\sigma(x,y,0)=\mathbf{U}^{n,s+1}(x,y),\end{cases} \tag{3.5}$$

where $\tilde{t}$ is a pseudo-time different from t, σ_{ij}^a can be found in (3.3), and we fix the moving grid I_{ij} in the RK stage $t^{n,s+1}$ $(s=0,1,2)$. Then the fully discrete Lagrangian type OFDG scheme using the third-order SSP-RK time stepping with the OF operator splitting strategy is as follows.

Stage 1:

$$\begin{aligned}
&\mathbf{x}^{n,1}_{i-\frac{1}{2},j-\frac{1}{2}} = \mathbf{x}^{n}_{i-\frac{1}{2},j-\frac{1}{2}} + \Delta t^n \mathbf{u}^{n}_{i-\frac{1}{2},j-\frac{1}{2}},\\
&\mathbf{A}^{n,1}\mathbf{U}^{n,1}_{c} = \mathbf{A}^{n}\mathbf{U}^{n}_{c} + \Delta t^n \mathbf{L}(\mathbf{U}^{n}_{c}),\\
&\mathbf{U}^{n,1}_{\sigma,c} = \mathcal{F}^{n,1}_{\mathrm{OF}}\mathbf{U}^{n,1}_{c};
\end{aligned} \tag{3.6}$$

Stage 2:

$$\begin{aligned}
&\mathbf{x}^{n,2}_{i-\frac{1}{2},j-\frac{1}{2}} = \frac{3}{4}\mathbf{x}^{n}_{i-\frac{1}{2},j-\frac{1}{2}} + \frac{1}{4}\mathbf{x}^{n,1}_{i-\frac{1}{2},j-\frac{1}{2}} + \frac{1}{4}\Delta t^n (\mathbf{u}_\sigma)^{n,1}_{i-\frac{1}{2},j-\frac{1}{2}},\\
&\mathbf{A}^{n,2}\mathbf{U}^{n,2}_{c} = \frac{3}{4}\mathbf{A}^{n}\mathbf{U}^{n}_{c} + \frac{1}{4}\mathbf{A}^{n,1}\mathbf{U}^{n,1}_{\sigma,c} + \frac{1}{4}\Delta t^n \mathbf{L}(\mathbf{U}^{n,1}_{\sigma,c}),\\
&\mathbf{U}^{n,2}_{\sigma,c} = \mathcal{F}^{n,2}_{\mathrm{OF}}\mathbf{U}^{n,2}_{c};
\end{aligned} \tag{3.7}$$

Stage 3:

$$\begin{aligned}
&\mathbf{x}^{n+1}_{i-\frac{1}{2},j-\frac{1}{2}} = \frac{1}{3}\mathbf{x}^{n}_{i-\frac{1}{2},j-\frac{1}{2}} + \frac{2}{3}\mathbf{x}^{n,2}_{i-\frac{1}{2},j-\frac{1}{2}} + \frac{2}{3}\Delta t^n (\mathbf{u}_\sigma)^{n,2}_{i-\frac{1}{2},j-\frac{1}{2}},\\
&\mathbf{A}^{n,3}\mathbf{U}^{n,3}_{c} = \frac{1}{3}\mathbf{A}^{n}\mathbf{U}^{n}_{c} + \frac{2}{3}\mathbf{A}^{n,2}\mathbf{U}^{n,2}_{\sigma,c} + \frac{2}{3}\Delta t^n \mathbf{L}(\mathbf{U}^{n,2}_{\sigma,c}),\\
&\mathbf{U}^{n,3}_{\sigma,c} = \mathcal{F}^{n,3}_{\mathrm{OF}}\mathbf{U}^{n,3}_{c},\\
&\mathbf{U}^{n+1}_{c} = \mathbf{U}^{n,3}_{\sigma,c};
\end{aligned} \tag{3.8}$$

where the operator $\mathcal{F}^{n,s+1}_{\mathrm{OF}}$ $(s = 0, 1, 2)$ is defined in (3.5). $\mathbf{L}$, $\mathbf{A}$, and $\mathbf{U}_c$ are defined in the formulas (2.19), (2.20), and (2.21). $\mathbf{U}_{\sigma,c}$ represents the coefficients of the numerical solution polynomials for the modified conserved variables $\mathbf{U}_\sigma$. We can find that the OF procedure is an extra addition after each RK stage, and does not interfere with the original Lagrangian type DG spatial discretization. Indeed, the fully discrete scheme (3.6)-(3.8) can be explained as the alternating evolution of the original semi-discrete Lagrangian type DG scheme (2.9)-(2.10) and the damping equation (3.5). Since the damping term is a high-order term for smooth solutions [31], the operator splitting strategy does not affect the original high order accuracy.

Next, we will explain how to solve the splitting damping equation (3.5) exactly.

3.3 Exact solver of the splitting damping equation

For the sake of simplicity, we take density as an example. Assume that we have obtained the numerical solution $\rho^{n,s+1}$ on the cell I_{ij} at the $(s+1)$-th RK time stage, which has the following expression:

$$\rho^{n,s+1} = \rho_c^{(0)}\phi_0 + \sum_{m=1}^{K} \rho_c^{(m)}\phi_m,$$

where $K = \frac{(k+1)(k+2)}{2} - 1$, $\{\phi_0,\ \phi_1,\ \cdots,\ \phi_K\}$ are the $K+1$ basis functions in the DG space V_h^k, and $\{\phi_1,\ \cdots,\ \phi_K\}$ are orthogonal to $\phi_0 = 1$. $\boldsymbol{\rho}_c = (\rho_c^{(0)},\ \rho_c^{(1)},\ \cdots,\ \rho_c^{(K)})^T$ represents the coefficients of the corresponding basis function. Our aim is to obtain the modified density $\rho_\sigma^{n,s+1}$ through the scheme (3.5). The modified density polynomial $\rho_\sigma^{n,s+1}$ is denoted as

$$\rho_\sigma^{n,s+1} = \rho_{\sigma,c}^{(0)}\phi_0 + \sum_{m=1}^{K} \rho_{\sigma,c}^{(m)}\phi_m,$$

where $\boldsymbol{\rho}_{\sigma,c} = (\rho_{\sigma,c}^{(0)},\ \rho_{\sigma,c}^{(1)},\ \cdots,\ \rho_{\sigma,c}^{(K)})^T$ represents the corresponding coefficients of the basis functions in the modified density $\rho_\sigma^{n,s+1}$.

Firstly, since $\iint_{I_{ij}^{n,s+1}} \left(\rho_\sigma - P_h^0 \rho_\sigma\right) dxdy = 0$, we have

$$\rho_{\sigma,c}^{(0)} = \rho_c^{(0)}.$$

In the following, let us consider the remaining coefficients. Denote

$$\boldsymbol{\rho}_c = (\rho_c^{(0)},\ \widetilde{\boldsymbol{\rho}}_c)^T,\ \boldsymbol{\rho}_{\sigma,c} = (\rho_{\sigma,c}^{(0)},\ \widetilde{\boldsymbol{\rho}}_{\sigma,c})^T,$$

where $\widetilde{\boldsymbol{\rho}}_c$ and $\widetilde{\boldsymbol{\rho}}_{\sigma,c}$ are the coefficients of the basis functions except the first basis function in $\rho^{n,s+1}$ and $\rho_\sigma^{n,s+1}$, respectively. Then the scheme (3.5) can be written as an ordinary differential equation (ODE) system:

$$\begin{cases} \dfrac{d}{dt}\widetilde{\boldsymbol{\rho}}_{\sigma,c} + \mathbf{B}\widetilde{\boldsymbol{\rho}}_{\sigma,c} = \mathbf{0}, \\ \widetilde{\boldsymbol{\rho}}_{\sigma,c}(0) = \widetilde{\boldsymbol{\rho}}_c, \end{cases} \tag{3.9}$$

where $\mathbf{B}$ is a $K \times K$ matrix. Since we are not using the completely orthogonal basis functions, the matrix $\mathbf{B}$ is not diagonal. For example, when $k = 2$, the matrix $\mathbf{B}$ is represented as follows:

$$\mathbf{B} = \frac{1}{h_{ij}} \begin{pmatrix} \sigma_{ij}^0 + \sigma_{ij}^1 & 0 & -\sigma_{ij}^2\widetilde{b}_{13} & -\sigma_{ij}^2\widetilde{b}_{14} & -\sigma_{ij}^2\widetilde{b}_{15} \\ 0 & \sigma_{ij}^0 + \sigma_{ij}^1 & -\sigma_{ij}^2\widetilde{b}_{23} & -\sigma_{ij}^2\widetilde{b}_{24} & -\sigma_{ij}^2\widetilde{b}_{25} \\ 0 & 0 & \sigma_{ij}^0 + \sigma_{ij}^1 + \sigma_{ij}^2 & 0 & 0 \\ 0 & 0 & 0 & \sigma_{ij}^0 + \sigma_{ij}^1 + \sigma_{ij}^2 & 0 \\ 0 & 0 & 0 & 0 & \sigma_{ij}^0 + \sigma_{ij}^1 + \sigma_{ij}^2 \end{pmatrix},$$

where

$$\begin{pmatrix} \widetilde{b}_{13} & \widetilde{b}_{14} & \widetilde{b}_{15} \\ \widetilde{b}_{23} & \widetilde{b}_{24} & \widetilde{b}_{25} \end{pmatrix} = \left(\iint_{I_{ij}^{n,s+1}} \begin{pmatrix} \phi_1\phi_1 & \phi_2\phi_1 \\ \phi_1\phi_2 & \phi_2\phi_2 \end{pmatrix} dxdy \right)^{-1} \left(\iint_{I_{ij}^{n,s+1}} \begin{pmatrix} \phi_3\phi_1 & \phi_4\phi_1 & \phi_5\phi_1 \\ \phi_3\phi_2 & \phi_4\phi_2 & \phi_5\phi_2 \end{pmatrix} dxdy \right).$$

Note that the ODE system (3.9) is linear, so its exact solution can be expressed explicitly. We assume that the matrices with eigenvalues and right eigenvectors of the matrix $\mathbf{B}$ are $\Lambda_{\mathbf{B}}$ and $\mathbf{R}_{\mathbf{B}}$, respectively. It is easy to obtain the exact solution of the ODE system (3.9) as

$$\widetilde{\boldsymbol{\rho}}_{\sigma,c}(\Delta t) = \mathbf{R}_{\mathbf{B}}\, e^{-\Delta t \Lambda_{\mathbf{B}}}\, \mathbf{R}_{\mathbf{B}}^{-1} \widetilde{\boldsymbol{\rho}}_c. \tag{3.10}$$

We can get the modified density polynomial as

$$\rho_\sigma^{n,s+1} = \rho_c^{(0)}\phi_0 + \sum_{m=1}^{K} \left(\mathbf{R}_{\mathbf{B}}\, e^{-\Delta t \Lambda_{\mathbf{B}}}\, \mathbf{R}_{\mathbf{B}}^{-1} \widetilde{\boldsymbol{\rho}}_c \right)^{(m)} \phi_m. \tag{3.11}$$

We can find that the modified density polynomial is obtained simply, efficiently, and accurately, thanks to the exact solver. The exact solvers of the splitting damping equation for other conserved variables such as momentum and total energy are similar to that of density, and we will not show the details.

In addition, it should be noted that the oscillation-free procedure could be applied only in those cells that may contain discontinuities, according to some discontinuity detector, which could reduce some computational cost. However, in this paper, we would like to verify the performance of the damping term when applied everywhere, in terms of oscillation control near discontinuities and accuracy in smooth regions. Also, since the OF procedure introduces only modest additional computational cost, we have decided to use it everywhere in the numerical tests.

4 Numerical tests

4.1 Accuracy tests

In this subsection, we will test the accuracy of the Lagrangian type DG scheme (2.9)-(2.10) and the Lagrangian type OFDG scheme (3.6)-(3.8) for the two-dimensional compressible Euler equations. The third order SSP Runge-Kutta time discretization is used. The Courant number λ is set to 0.5, 0.3, and 0.1 respectively when $k = 1$, 2 and 3. For the accuracy tests with $k = 3$, we also replace Δl_{ij} by $(\Delta l_{ij})^{4/3}$ in order to make the temporal and spatial orders match.

Example 4.1 We first consider an isentropic problem with the periodic boundary condition. The initial condition is:

$$\rho_0(x, y) = \frac{1 + 0.2\sin(\frac{x+y}{2})}{\sqrt{6}},\ u_0(x, y) = v_0(x, y) = \sqrt{\frac{\gamma}{2}}\rho_0,\ p_0(x, y) = \rho_0^{\gamma},\ (x, y) \in [0, 4\pi]^2.$$

We set $\gamma = 3$, and we can verify that $\sqrt{6}\rho(x, y, t)$ is the exact solution of the two-dimensional Burgers' equation:

$$\beta_t + \left(\frac{\beta^2}{2}\right)_x + \left(\frac{\beta^2}{2}\right)_y = 0, \quad \beta_0(x, y) = 1 + 0.2\sin\left(\frac{x + y}{2}\right),$$

and the exact solutions (u, v) and p satisfy:

$$u(x, y, t) = v(x, y, t) = \sqrt{\frac{\gamma}{2}}\rho(x, y, t),\ \ p(x, y, t) = \rho(x, y, t)^{\gamma}.$$

The initial computational domain is divided into uniform squares with mesh size $h = \frac{4\pi}{N}$. The final time is $t = 1$.

The errors and numerical rates of convergence are shown in Tables 1, 2, 3, where we can see the $(k + 1)$-th order accuracy of the Lagrangian type DG scheme (2.9)-(2.10) and the Lagrangian type OFDG scheme (3.6)-(3.8) for the density, momentum and energy. This indicates that our DG schemes have the optimal order accuracy, and the damping term does not pollute the order of accuracy.

Example 4.2 **Taylor-Green vortex problem**

The two-dimensional Taylor-Green vortex problem [2] is a benchmark test with analytical solutions. The initial condition is

$$\rho_0 = 1,\ \ p_0(x, y) = \frac{1}{4}(\cos(2\pi x) + \cos(2\pi y)) + 1,$$
$$u_0(x, y) = \sin(\pi x)\cos(\pi y),\ \ v_0(x, y) = -\cos(\pi x)\sin(\pi y).$$

The working fluid is ideal gas with $\gamma = \frac{5}{3}$. In the compressible inviscid case, we need a source term in the energy equation to make the flow steady state:

$$S = \frac{\pi}{4(\gamma - 1)}(\cos(3\pi x)\cos(\pi y) - \cos(\pi x)\cos(3\pi y)).$$

The computational domain is $[0, 1] \times [0, 1]$. We set the final time $t = 0.1$.

The numerical errors are listed in Tables 4, 5, 6. From the tables, we can find the $(k + 1)$-th order accuracy for the density, momentum and energy. Therefore, we observe again that both the Lagrangian type DG scheme (2.9)-(2.10) and the Lagrangian type OFDG scheme (3.6)-(3.8) can achieve the optimal order accuracy.

Table 1 The errors and orders of accuracy for the density solutions in Example 4.1.

		N(=M)	L^2 error	Order(L^2)	L^1 error	Order(L^1)	L^∞ error	Order(L^∞)
DG	$\mathcal{P}^1$	20	8.2375E-04	-	5.8575E-04	–	5.0678E-03	–
		40	2.0973E-04	1.97	1.4606E-04	2.00	1.3440E-03	1.92
		80	5.2897E-05	1.99	3.6523E-05	2.00	3.4674E-04	1.96
		160	1.3280E-05	1.99	9.1358E-06	2.00	8.7992E-05	1.98
	$\mathcal{P}^2$	20	6.1046E-05	–	3.9122E-05	–	6.8252E-04	–
		40	8.0092E-06	2.93	5.0641E-06	2.95	9.6179E-05	2.83
		80	1.0242E-06	2.97	6.4198E-07	2.98	1.2572E-05	2.94
		160	1.2940E-07	2.99	8.0720E-08	2.99	1.6071E-06	2.97
	$\mathcal{P}^3$	20	4.8608E-06	–	2.7900E-06	–	7.1200E-05	–
		40	3.2961E-07	3.88	1.8102E-07	3.95	4.8477E-06	3.88
		80	2.1416E-08	3.94	1.1514E-08	3.98	3.2101E-07	3.92
		160	1.3641E-09	3.97	7.2629E-10	3.99	2.0765E-08	3.95
OFDG	$\mathcal{P}^1$	20	9.4821E-04	–	7.1325E-04	–	5.3302E-03	-
		40	2.2936E-04	2.05	1.6570E-04	2.11	1.4009E-03	1.93
		80	5.5537E-05	2.05	3.9167E-05	2.08	3.5735E-04	1.97
		160	1.3617E-05	2.03	9.4703E-06	2.05	8.9527E-05	2.00
	$\mathcal{P}^2$	20	7.1767E-05	–	4.7066E-05	–	7.5197E-04	–
		40	8.6827E-06	3.05	5.5091E-06	3.10	1.0358E-04	2.86
		80	1.0630E-06	3.03	6.6668E-07	3.05	1.3125E-05	2.98
		160	1.3167E-07	3.01	8.2098E-08	3.02	1.6416E-06	3.00
	$\mathcal{P}^3$	20	7.3743E-06	–	4.4571E-06	–	9.5763E-05	–
		40	3.8476E-07	4.26	2.1879E-07	4.35	6.0408E-06	3.99
		80	2.2326E-08	4.12	1.2290E-08	4.15	3.5948E-07	4.07
		160	1.3791E-09	4.02	7.4387E-10	4.05	2.2046E-08	4.03

Table 2 The errors and orders of accuracy for the momentum solutions in Example 4.1.

		N(=M)	L^2 error	Order(L^2)	L^1 error	Order(L^1)	L^∞ error	Order(L^∞)
DG	$\mathcal{P}^1$	20	1.0568E-03	–	7.9239E-04	–	6.2755E-03	–
		40	2.6353E-04	2.00	1.9593E-04	2.02	1.6496E-03	1.93
		80	6.5694E-05	2.00	4.8638E-05	2.01	4.1683E-04	1.99
		160	1.6399E-05	2.00	1.2123E-05	2.00	1.0469E-04	1.99
	$\mathcal{P}^2$	20	8.4918E-05	–	5.5959E-05	–	1.0036E-03	–
		40	1.1021E-05	2.95	7.1552E-06	2.97	1.3764E-04	2.87
		80	1.3993E-06	2.98	9.0265E-07	2.99	1.7733E-05	2.96
		160	1.7611E-07	2.99	1.1329E-07	2.99	2.2492E-06	2.98
	$\mathcal{P}^3$	20	6.7764E-06	–	4.1285E-06	–	9.8943E-05	–
		40	4.5850E-07	3.89	2.7021E-07	3.93	7.1571E-06	3.79
		80	2.9717E-08	3.95	1.7183E-08	3.98	4.8068E-07	3.90
		160	1.8901E-09	3.98	1.0844E-09	3.99	3.1152E-08	3.95
OFDG	$\mathcal{P}^1$	20	1.4913E-03	–	1.2239E-03	–	7.1205E-03	–
		40	3.7737E-04	1.98	3.0797E-04	1.99	1.8861E-03	1.92
		80	9.4138E-05	2.00	7.6684E-05	2.01	4.8373E-04	1.96
		160	2.3476E-05	2.00	1.9103E-05	2.01	1.2161E-04	1.99
	$\mathcal{P}^2$	20	1.1954E-04	–	9.2127E-05	–	8.7016E-04	–
		40	1.4668E-05	3.03	1.1209E-05	3.04	1.2377E-04	2.81
		80	1.8158E-06	3.01	1.3875E-06	3.01	1.5589E-05	2.99
		160	2.2614E-07	3.01	1.7281E-07	3.01	1.9388E-06	3.01
	$\mathcal{P}^3$	20	1.1580E-05	–	8.0404E-06	–	1.0556E-04	–
		40	6.8301E-07	4.08	4.6762E-07	4.10	7.2332E-06	3.87
		80	4.2371E-08	4.01	2.8958E-08	4.01	4.5620E-07	3.99
		160	2.6683E-09	3.99	1.8188E-09	3.99	2.8426E-08	4.00

Table 3 The errors and orders of accuracy for the energy solutions in Example 4.1.

		N(=M)	L^2 error	Order(L^2)	L^1 error	Order(L^1)	L^∞ error	Order(L^∞)
DG	$\mathcal{P}^1$	20	7.3148E-04	–	5.2874E-04	–	5.1213E-03	–
		40	1.7889E-04	2.03	1.2635E-04	2.07	1.3272E-03	1.95
		80	4.4097E-05	2.02	3.0710E-05	2.04	3.3499E-04	1.99
		160	1.0947E-05	2.01	7.5661E-06	2.02	8.3929E-05	2.00
	$\mathcal{P}^2$	20	6.0377E-05	–	3.7284E-05	–	7.7200E-04	–
		40	7.7335E-06	2.97	4.7159E-06	2.98	1.0119E-04	2.93
		80	9.7428E-07	2.99	5.9121E-07	3.00	1.3030E-05	2.96
		160	1.2210E-07	3.00	7.3890E-08	3.00	1.6403E-06	2.99
	$\mathcal{P}^3$	20	4.6402E-06	–	2.8076E-06	–	7.2304E-05	–
		40	3.0729E-07	3.92	1.7954E-07	3.97	5.0570E-06	3.84
		80	1.9690E-08	3.96	1.1314E-08	3.99	3.3395E-07	3.92
		160	1.2449E-09	3.98	7.1046E-10	3.99	2.1398E-08	3.96
OFDG	$\mathcal{P}^1$	20	9.7330E-04	–	7.5344E-04	–	4.7661E-03	–
		40	2.3294E-04	2.06	1.7545E-04	2.10	1.2677E-03	1.91
		80	5.6013E-05	2.06	4.1476E-05	2.08	3.2050E-04	1.98
		160	1.3691E-05	2.03	1.0032E-05	2.05	8.0121E-05	2.00
	$\mathcal{P}^2$	20	7.6868E-05	–	5.4810E-05	–	6.5052E-04	–
		40	9.2689E-06	3.05	6.5458E-06	3.07	8.9418E-05	2.86
		80	1.1399E-06	3.02	8.0145E-07	3.03	1.1184E-05	3.00
		160	1.4162E-07	3.01	9.9333E-08	3.01	1.3895E-06	3.01
	$\mathcal{P}^3$	20	7.1878E-06	–	4.5471E-06	–	7.0220E-05	–
		40	4.1143E-07	4.13	2.5067E-07	4.18	4.9077E-06	3.84
		80	2.5137E-08	4.03	1.5062E-08	4.06	3.0734E-07	4.00
		160	1.5719E-09	4.00	9.3559E-10	4.01	1.9185E-08	4.00

Table 4 The errors and orders of accuracy for the density solutions in Example 4.2.

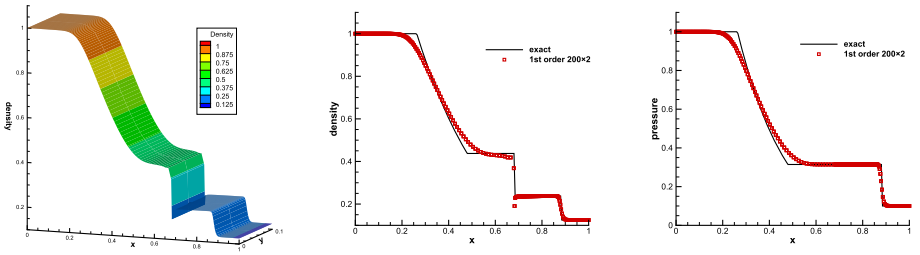
		N(=M)	L^2 error	Order(L^2)	L^1 error	Order(L^1)	L^∞ error	Order(L^∞)
DG	$\mathcal{P}^1$	20	6.4910E-04	–	5.1353E-04	–	2.1965E-03	–
		40	1.6699E-04	1.96	1.3104E-04	1.97	5.5744E-04	1.98
		80	4.2355E-05	1.98	3.3099E-05	1.99	1.4005E-04	1.99
		160	1.0666E-05	1.99	8.3204E-06	1.99	3.5093E-05	2.00
	$\mathcal{P}^2$	20	2.5506E-05	–	1.7983E-05	–	1.6549E-04	–
		40	3.3248E-06	2.94	2.3465E-06	2.94	2.1401E-05	2.95
		80	4.2413E-07	2.97	2.9941E-07	2.97	2.7255E-06	2.98
		160	5.3546E-08	2.99	3.7810E-08	2.99	3.4360E-07	2.99
	$\mathcal{P}^3$	20	7.8615E-07	–	5.3832E-07	–	6.6927E-06	–
		40	5.0821E-08	3.95	3.4578E-08	3.96	4.3883E-07	3.93
		80	3.2245E-09	3.98	2.1879E-09	3.98	2.7541E-08	3.99
		160	2.1079E-10	3.94	1.4695E-10	3.90	2.1175E-09	3.70
OFDG	$\mathcal{P}^1$	20	4.2550E-03	–	3.0999E-03	-	1.7860E-02	–
		40	1.3739E-03	1.63	9.8640E-04	1.65	6.1676E-03	1.53
		80	3.9677E-04	1.79	2.8174E-04	1.81	1.8629E-03	1.73
		160	1.0728E-04	1.89	7.5606E-05	1.90	5.1648E-04	1.85
	$\mathcal{P}^2$	20	1.4731E-04	–	1.0428E-04	–	8.0711E-04	–
		40	1.7733E-05	3.05	1.2907E-05	3.01	9.4033E-05	3.10
		80	2.1731E-06	3.03	1.6078E-06	3.01	1.1269E-05	3.06
		160	2.6939E-07	3.01	2.0097E-07	3.00	1.3816E-06	3.03
	$\mathcal{P}^3$	20	5.6699E-06	–	3.9461E-06	–	3.6595E-05	–
		40	3.7344E-07	3.92	2.5986E-07	3.93	2.3263E-06	3.98
		80	2.4231E-08	3.95	1.6849E-08	3.95	1.4438E-07	4.01
		160	1.5472E-09	3.97	1.0781E-09	3.97	9.1022E-09	3.99

Table 5 The errors and orders of accuracy for the momentum solutions in Example 4.2.

		N(=M)	L^2 error	Order(L^2)	L^1 error	Order(L^1)	L^∞ error	Order(L^∞)
DG	$\mathcal{P}^1$	20	2.2706E-03	–	1.9030E-03	–	8.6506E-03	–
		40	5.7151E-04	1.99	4.7812E-04	1.99	2.1736E-03	1.99
		80	1.4335E-04	2.00	1.1983E-04	2.00	5.4498E-04	2.00
		160	3.5897E-05	2.00	2.9994E-05	2.00	1.3639E-04	2.00
	$\mathcal{P}^2$	20	6.8562E-05	–	5.6550E-05	–	3.5204E-04	–
		40	8.5418E-06	3.00	7.0658E-06	3.00	4.3457E-05	3.02
		80	1.0665E-06	3.00	8.8360E-07	3.00	5.3956E-06	3.01
		160	1.3323E-07	3.00	1.1046E-07	3.00	6.7191E-07	3.01
	$\mathcal{P}^3$	20	2.4436E-06	–	1.9438E-06	–	1.8580E-05	–
		40	1.5336E-07	3.99	1.2212E-07	3.99	1.1664E-06	3.99
		80	9.6006E-09	4.00	7.6455E-09	4.00	7.2455E-08	4.01
		160	6.0163E-10	4.00	4.7953E-10	4.00	4.5210E-09	4.00
OFDG	$\mathcal{P}^1$	20	1.1144E-02	–	7.7647E-03	–	5.1481E-02	–
		40	3.5403E-03	1.65	2.3276E-03	1.74	1.8309E-02	1.49
		80	1.0315E-03	1.78	6.5082E-04	1.84	5.7659E-03	1.67
		160	2.8201E-04	1.87	1.7343E-04	1.91	1.6538E-03	1.80
	$\mathcal{P}^2$	20	3.1751E-04	–	2.2794E-04	–	1.1691E-03	–
		40	3.7292E-05	3.09	2.7109E-05	3.07	1.4046E-04	3.06
		80	4.5607E-06	3.03	3.3212E-06	3.03	1.7517E-05	3.00
		160	5.6574E-07	3.01	4.1193E-07	3.01	2.1866E-06	3.00
	$\mathcal{P}^3$	20	1.3063E-05	–	8.5350E-06	–	8.4116E-05	–
		40	9.1536E-07	3.84	5.6475E-07	3.92	6.3294E-06	3.73
		80	6.0597E-08	3.92	3.6472E-08	3.95	4.2734E-07	4.00
		160	3.8957E-09	3.96	2.3213E-09	3.97	2.7633E-08	3.95

Table 6 The errors and orders of accuracy for the energy solutions in Example 4.2.

		N(=M)	L^2 error	Order(L^2)	L^1 error	Order(L^1)	L^∞ error	Order(L^∞)
DG	$\mathcal{P}^1$	20	2.2955E-03	–	1.8034E-03	–	8.5865E-03	–
		40	5.7895E-04	1.99	4.5262E-04	1.99	2.2278E-03	1.95
		80	1.4542E-04	1.99	1.1342E-04	2.00	5.6690E-04	1.97
		160	3.6446E-05	2.00	2.8390E-05	2.00	1.4287E-04	1.99
	$\mathcal{P}^2$	20	8.1686E-05	–	6.0902E-05	–	5.7556E-04	–
		40	1.0280E-05	2.99	7.6341E-06	3.00	7.3266E-05	2.97
		80	1.2891E-06	3.00	9.5518E-07	3.00	9.2078E-06	2.99
		160	1.6138E-07	3.00	1.1945E-07	3.00	1.1544E-06	3.00
	$\mathcal{P}^3$	20	2.7562E-06	–	2.0330E-06	–	2.0444E-05	–
		40	1.7276E-07	4.00	1.2742E-07	4.00	1.3053E-06	3.97
		80	1.0814E-08	4.00	7.9604E-09	4.00	8.2110E-08	3.99
		160	6.8540E-10	3.98	5.0454E-10	3.98	5.9317E-09	3.79
OFDG	$\mathcal{P}^1$	20	5.0800E-03	–	3.8395E-03	–	2.1364E-02	–
		40	1.3640E-03	1.90	1.0028E-03	1.94	6.6510E-03	1.68
		80	3.6222E-04	1.91	2.6181E-04	1.94	1.9064E-03	1.80
		160	9.4155E-05	1.94	6.7384E-05	1.96	5.1628E-04	1.89
	$\mathcal{P}^2$	20	1.5115E-04	–	1.1837E-04	–	4.9682E-04	–
		40	1.7273E-05	3.13	1.3440E-05	3.14	5.4844E-05	3.18
		80	2.0914E-06	3.05	1.6209E-06	3.05	6.4416E-06	3.09
		160	2.5836E-07	3.02	1.9996E-07	3.02	7.8252E-07	3.04
	$\mathcal{P}^3$	20	5.3762E-06	–	4.0490E-06	–	2.5630E-05	–
		40	3.3737E-07	3.99	2.4438E-07	4.05	1.9693E-06	3.70
		80	2.1490E-08	3.97	1.5292E-08	4.00	1.3508E-07	3.87
		160	1.3667E-09	3.98	9.6751E-10	3.98	8.9204E-09	3.92



(a) density (b) density profile (c) pressure profile

Fig. 1 The results of the Sod problem with the $\mathcal{P}^0$ DG scheme at $t = 0.2$.

4.2 Non-oscillatory tests

In this subsection, we will verify the non-oscillatory and high-resolution performance of our designed DG schemes. At $t = 0$, we set the cell edges to align with the initial discontinuity, so that there are no mixed cells. For the case of $k \leq 1$, the flux term $\int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g) \cdot \mathbf{n}\right) ds$ in the cell average equations is discarded directly, as described in [15]. For the case of $k \geq 2$, the flux term $\int_{\partial I_{ij}(t)} \left(\widehat{\mathbf{U}}(\widehat{\mathbf{u}} - \mathbf{u}_g) \cdot \mathbf{n}\right) ds$ in the cell average equations is not calculated on the specific cell edges, as detailed in the subsection 2.3. These specific cell edges include contact discontinuities or material interfaces (marked as the logical number of the cell edges located at the initial discontinuity at $t = 0$), as well as shocks marked by the KXRCF detection technique.

We will test nine well-known examples of the Euler equations on the general quadrilateral grids or polar grids, including the one-dimensional Riemann problems (the Sod, Lax, blast wave and Shu-Osher problems) on a two-dimensional grid, the two-dimensional polar Riemann problems (the polar Sod, polar Shu-Osher and Kidder isentropic compression problems), and the classical two-dimensional Lagrangian examples (the Sedov and Saltzman problems).

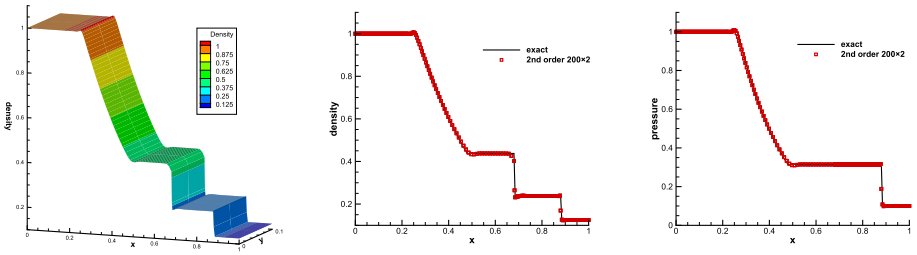
Example 4.3 The multi-material Sod shock tube problem

The Sod shock tube is a classical test problem. We set the initial condition as follows:

$$(\rho_0,\ u_0,\ v_0,\ p_0,\ \gamma) = \begin{cases} (1,\ 0,\ 0,\ 1,\ 1.4), & 0 \leq x < 0.5, \\ (0.125,\ 0,\ 0,\ 0.1,\ 5/3), & 0.5 \leq x < 1. \end{cases}$$

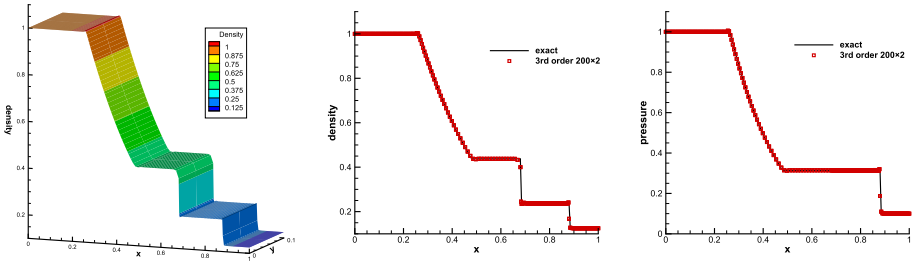
The materials have different ratio of specific heats γ. The computational domain is $[0, 1] \times [0, 0.1]$. The left and right boundary conditions are prescribed by the initial conditions, and the reflective boundary conditions are applied on both the upper and lower boundaries. This problem can be explained as running a one-dimensional Sod shock tube on a two-dimensional grid. The initial grid consists of 200×2 uniform rectangular cells, where the interface coincides with the initial discontinuity and there are no mixed cells.

Figs. 1, 2, 3 show the numerical solutions of the $\mathcal{P}^k$ polynomials at $t = 0.2$, where we observe that the numerical solutions almost agree with the analytical solutions when $k \geq 1$. Moreover, in Fig. 4, we compare the results of the first-order, second-order and third-order DG schemes on a 50×2 grid. From the figures, we find that the results of the third-order scheme perform better than those of the first-order and second-order schemes.



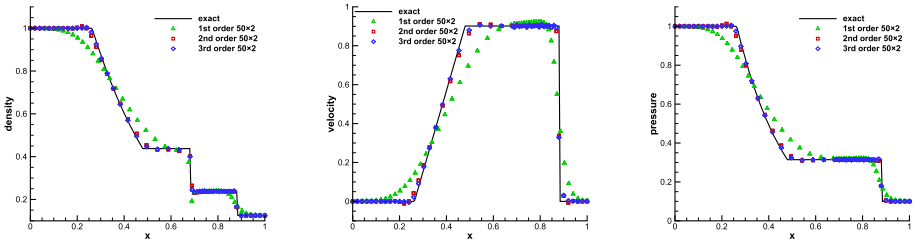
(a) density (b) density profile (c) pressure profile

Fig. 2 The results of the Sod problem with the $\mathcal{P}^1$ DG scheme at $t = 0.2$.



(a) density (b) density profile (c) pressure profile

Fig. 3 The results of the Sod problem with the $\mathcal{P}^2$ DG scheme at $t = 0.2$.



(a) density (b) velocity (c) pressure

Fig. 4 The results of the Sod problem on a 50×2 grid.

Example 4.4 The Lax problem

The Lax problem is another well-known Riemann problem. The initial condition is as follows.

$$(\rho_0,\ u_0,\ v_0,\ p_0) = \begin{cases} (0.445,\ 0.698,\ 0,\ 3.528), & 5 \le x < 10, \\ (0.5,\ 0,\ 0,\ 0.571), & 10 \le x < 15. \end{cases}$$

The initial domain is $[5, 15] \times [0, 0.1]$ and $\gamma = 1.4$. Initial values are fixed on the left and right boundaries, and the reflective boundary conditions are imposed on the lower and upper boundaries.

We run the Lax problem on a 100×2 grid, where the interface coincides with the discontinuity. In Fig. 5, we plot the density contour at $t = 1$. In Fig. 6, we compare the density

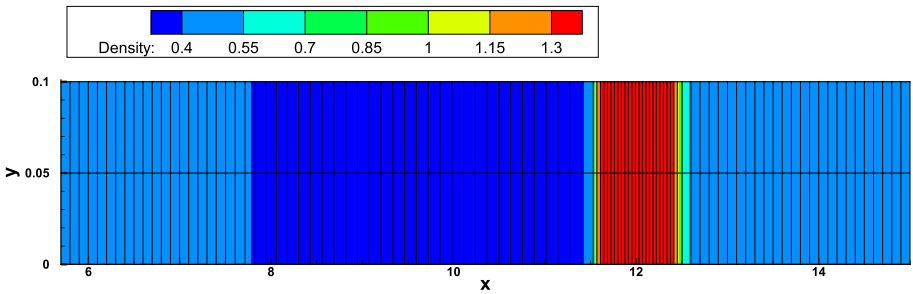


Fig. 5 The density of the Lax problem with the $\mathcal{P}^2$ OFDG scheme on a 100×2 grid at $t = 1$.

profiles of the Lagrangian type DG scheme with and without the OF procedure. It shows that the OFDG scheme can effectively control spurious oscillations.

Example 4.5 The interaction of blast wave problem

We consider the interaction of blast waves of the Euler equations. The initial data is

$$\rho_0 = 1,\ u_0 = 0,\ v_0 = 0,\ p_0 = \begin{cases} 10^3, & 0 < x < 0.1, \\ 10^{-2}, & 0.1 < x < 0.9, \\ 10^2, & 0.9 < x < 1.0. \end{cases}$$

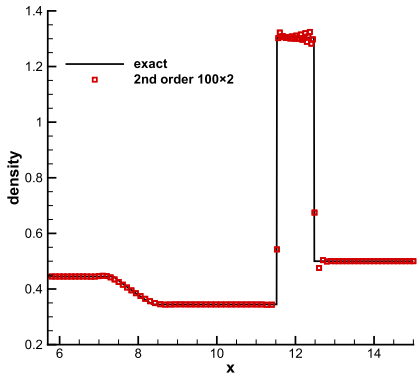
The computational domain is $[0, 1] \times [0, 0.01]$ filled with an ideal gas with $\gamma = 1.4$. The reflective boundary condition is used on all boundaries. The final time is $t = 0.038$. The grid consists of 400×2 cells. The initial grid is uniform.

The results using the OFDG scheme are shown in Figs. 7-8, where characteristic decomposition is used in the damping coefficients. The reference solution is computed by a fifth order Eulerian WENO scheme with 16000 grid points. We can see no spurious oscillations near the discontinuities, even in local zoomed-in pictures. We also notice that some overshoots appear near the discontinuity such as the contact wave. Such overshoots are caused by the Lagrangian framework, since they are already present for the first-order scheme, which have been explained in [5, 7].

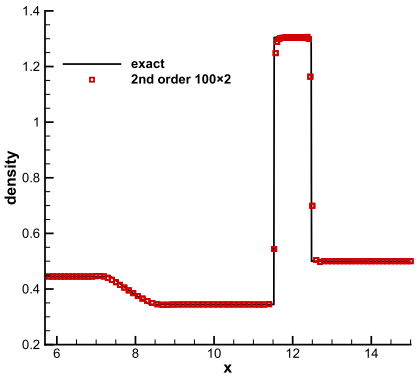
For comparison, we also test the blast wave problem using the OFDG scheme without characteristic decomposition. The numerical solutions are shown in Fig. 9, where we observe that the results are much less satisfactory, so characteristic decomposition is essential in the damping coefficients of the OFDG scheme.

Example 4.6 The Shu-Osher problem

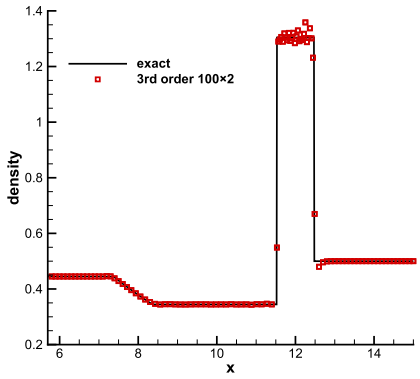
The Shu-Osher problem [41] is a typical example describing the interaction between a right-moving Mach=3 shock and sine waves. Due to the coexistence of shocks and the complicated smooth structures, this problem is very suitable for examining the capability of high order schemes. The initial computational domain is $[-10, 5] \times [0, 0.1]$. The initial condition is given as



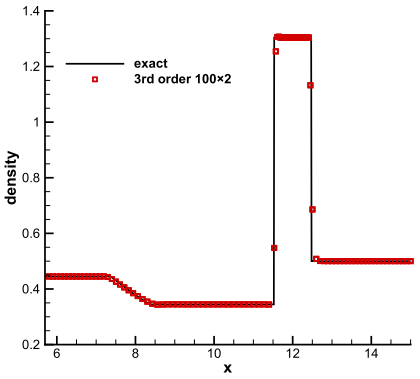
(a) $\mathcal{P}^1$ density (DG)



(b) $\mathcal{P}^1$ density (OFDG)



(c) $\mathcal{P}^2$ density (DG)



(d) $\mathcal{P}^2$ density (OFDG)

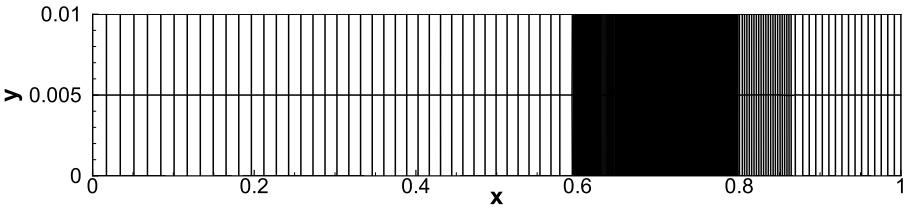
Fig. 6 The results of the Lax problem without or with the OF procedure.

$$(\rho_0,\ u_0,\ v_0,\ p_0) = \begin{cases} (3.857143,\ 2.629369,\ 0,\ 10.33333), & x < -4, \\ (1 + 0.2\sin(5x),\ 0,\ 0,\ 1), & x \geq -4. \end{cases}$$

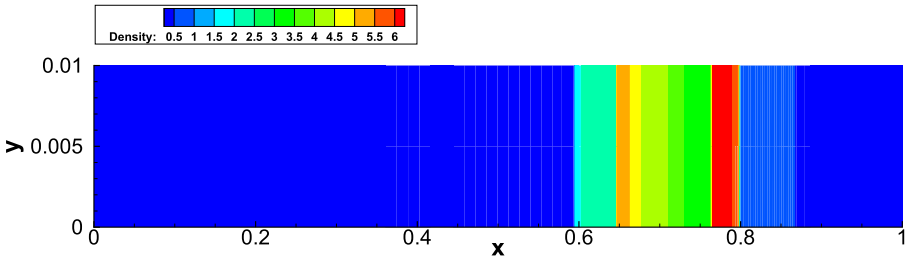
The final time is $t = 1.8$. The ratio of specific heat capacities $\gamma = 1.4$.

The results with the $\mathcal{P}^1$ or $\mathcal{P}^2$ OFDG schemes are shown in Figs. 10-11. The reference solution is obtained by the fifth-order Eulerian WENO scheme with 2000 grid points. From the figures, we find that the density profile of the third-order OFDG scheme with 180 cells is similar to that of the second-order OFDG scheme with 300 cells. Here the true degree of freedom for the second-order scheme is 600, while the degree of freedom for the third-order scheme is only 540, indicating that the higher order OFDG schemes use fewer degrees of freedom with the same resolution. We also noticed some undershoots near the contact discontinuity, which are caused by the Lagrangian framework [5, 7].

In the following three examples, we will test the performance of the Lagrangian type DG scheme on the initially equal-angled polar grids. $r = \sqrt{x^2 + y^2}$ is the radial coordinate. Reflective boundary conditions are applied to the x and y axes.



(a) mesh



(b) density

Fig. 7 The results of the blast wave problem with the $\mathcal{P}^1$ OFDG scheme on a 400×2 grid at $t = 0.038$.

Example 4.7 The multi-material polar Sod shock tube problem

We consider the extension of the classical Sod shock tube to the case of the cylindrical geometry. The computational domain is $[0, 1] \times [0, \pi/2]$ defined in the polar coordinates. The initial condition is

$$(\rho_0,\ u_0,\ v_0,\ p_0,\ \gamma) = \begin{cases} (1,\ 0,\ 0,\ 1,\ 1.4), & 0 \le r < 0.5, \\ (0.125,\ 0,\ 0,\ 0.1,\ 5/3), & 0.5 \le r < 1. \end{cases}$$

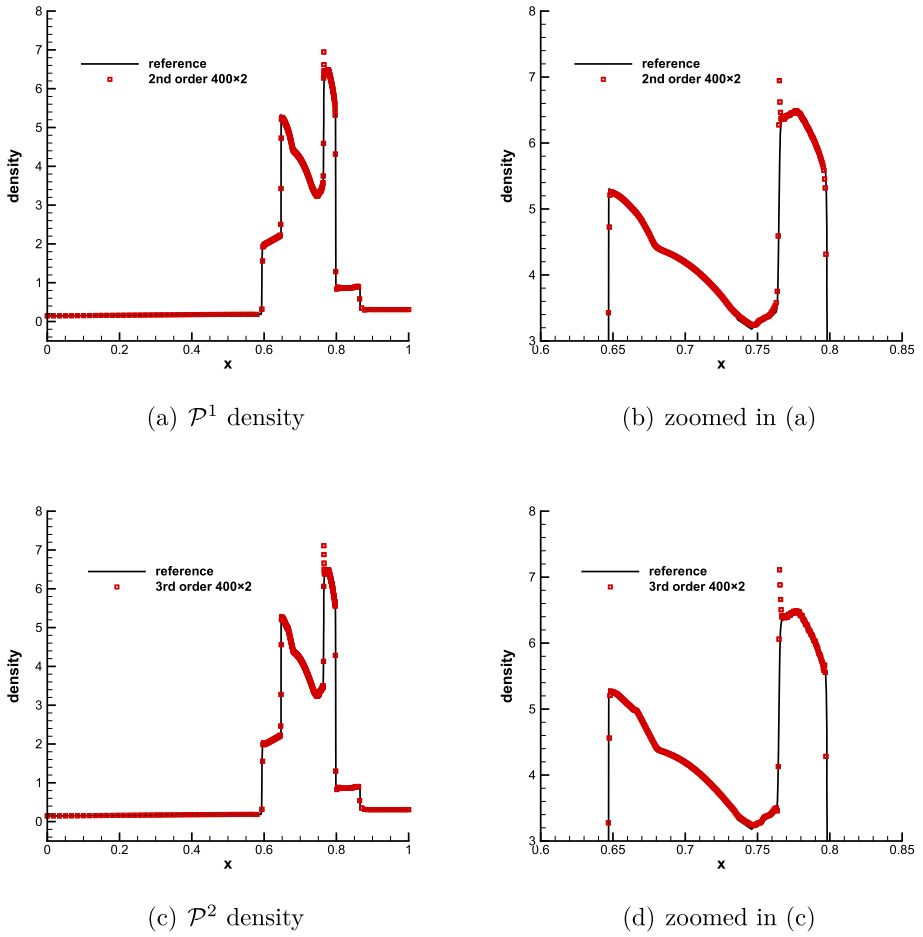
The wall boundary condition is applied on the outer boundary ($r = 1$). The initial grid consists of 150×20 equal-angled polar cells, where the interface coincides with the discontinuity and there are no mixed cells. The final time is set to $t = 0.2$.

The results with the $\mathcal{P}^2$ DG scheme (2.9)-(2.10) are shown in Fig. 12, including the mesh, density contour, and density as a function of the radial radius at all grid points. Here the reference solution is obtained by one-dimensional second-order Lagrangian code in the cylindrical coordinate with 1000 cells. We can observe the good radial symmetry of the Lagrangian type DG scheme.

Example 4.8 The polar Shu-Osher problem

We consider the Shu-Osher problem in the polar coordinates. The initial domain is $[1, 16] \times [0, \pi/2]$ defined in the polar coordinates. The initial condition is described as

$$(\rho_0,\ u_0,\ v_0,\ p_0) = \begin{cases} (3.857143,\ 2.629369x/r,\ 2.629369y/r,\ 10.33333), & r < 7, \\ (1 + 0.2\sin(5r),\ 0,\ 0,\ 1), & r \ge 7. \end{cases}$$



(a) $\mathcal{P}^1$ density (b) zoomed in (a)

(c) $\mathcal{P}^2$ density (d) zoomed in (c)

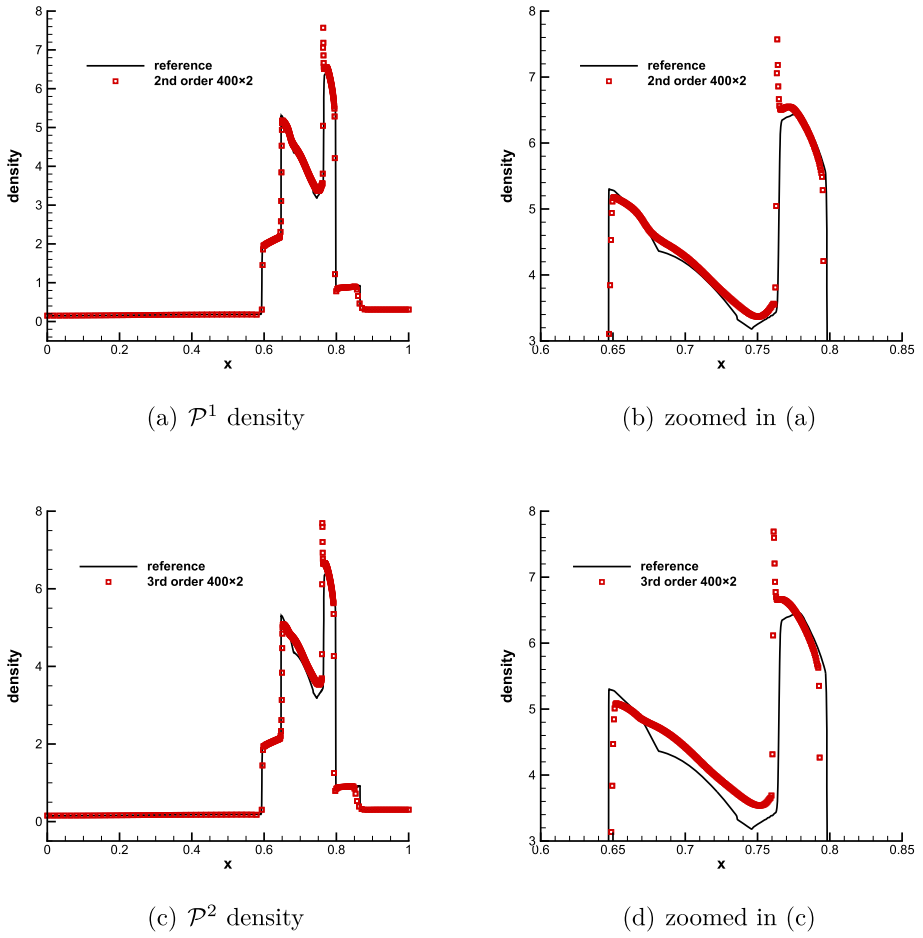
Fig. 8 The results of the blast wave problem using the OFDG scheme with characteristic decomposition.

The final time is $t = 1.8$.

We test this problem using the $\mathcal{P}^2$ Lagrangian type OFDG scheme on an initially equal-angled polar grid of 120×15 cells. The results are shown in Fig. 13, including the mesh, density contour, and density versus radial radius r at all grid points. Here the reference solution is from a one-dimensional third-order Lagrangian code in the cylindrical coordinate with 2000 cells. In the plots of grid and density contour, we observe the good radial symmetry. The non-oscillatory property is well reflected in the scatter plot of density. Moreover, we compare the density solutions of second-order and third-order OFDG schemes. The results are shown in Fig. 14. We find that the fine structures in the density profiles make the third-order scheme perform much better than the second-order scheme.

Example 4.9 **Kidder's isentropic compression problem**

In [21], Kidder computed the analytical solution of the self-similar isentropic compression for a shell filled with perfect gas. At $t = 0$, we set the computational region as $[r_1, r_2] \times$



(a) $\mathcal{P}^1$ density (b) zoomed in (a)

(c) $\mathcal{P}^2$ density (d) zoomed in (c)

Fig. 9 The results of the blast wave problem using the OFDG scheme without characteristic decomposition.

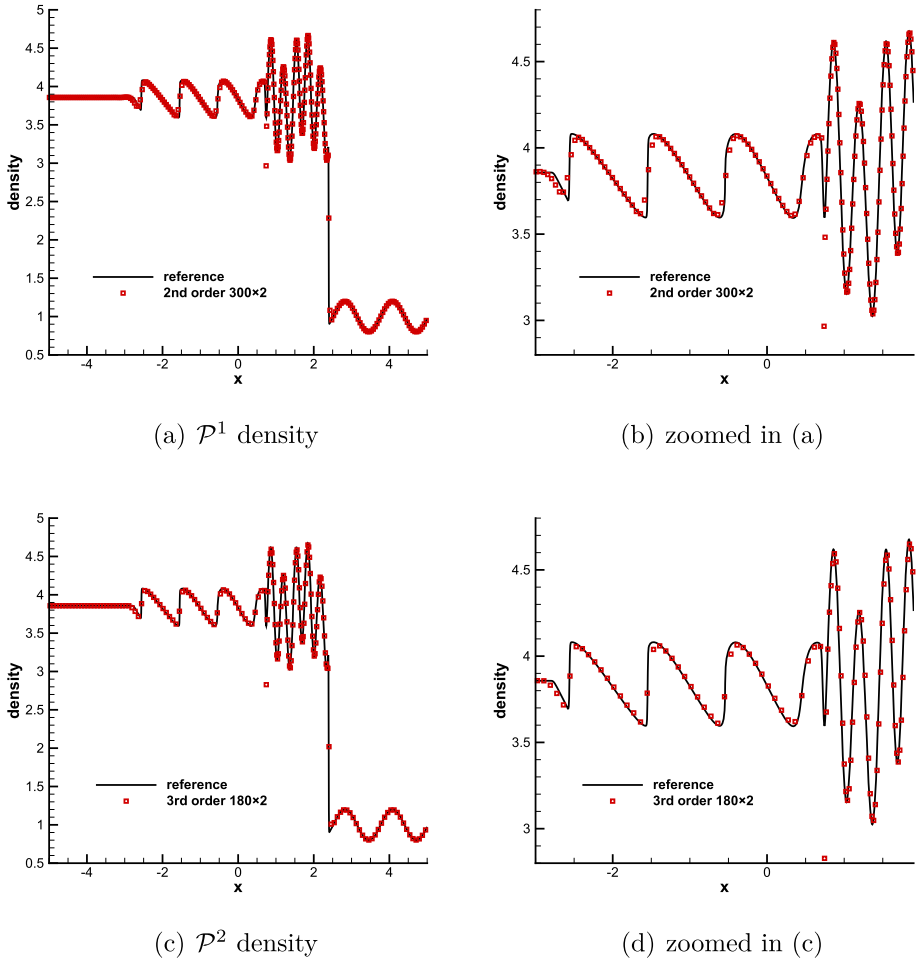
$[0, \pi/2]$ in the polar coordinates. The initial density, velocity and pressure are expressed as follows,

$$\rho_0(r) = \left(\frac{r_2^2 - r^2}{r_2^2 - r_1^2} \rho_1^{\gamma-1} + \frac{r^2 - r_1^2}{r_2^2 - r_1^2} \rho_2^{\gamma-1} \right)^{\frac{1}{\gamma-1}}, \quad u_0 = v_0 = 0, \quad p_0(r) = s\,(\rho_0(\mu))^{\gamma},$$

where $r_1 = 0.9$, $r_2 = 1.0$, $\rho_1 = 10^{-3}$, $\rho_2 = 10^{-2}$, $s = 10^5$, and $\gamma = 2$. Moreover, the pressure $p_1(t)$ and $p_2(t)$ are imposed continuously at the internal and external boundaries, respectively, and their expressions are as follows

$$p_1(t) = p_1^0 a(t)^{-\frac{2\gamma}{\gamma-1}}, \quad p_2(t) = p_2^0 a(t)^{-\frac{2\gamma}{\gamma-1}},$$

where $p_1^0 = 0.1$, $p_2^0 = 10$ and $a(t) = \sqrt{1 - \left(\frac{t}{\tau}\right)^2}$. $\tau = 7.265 \times 10^{-3}$ is the focusing time of the shell and $t \in [0, \tau)$ is the evolving time. The analytical solutions for this problem in cylindrical geometry are as follows:



(a) $\mathcal{P}^1$ density (b) zoomed in (a)

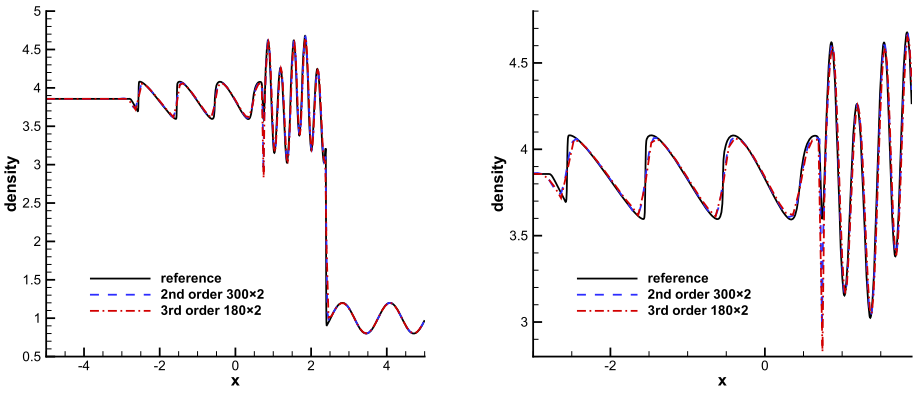
(c) $\mathcal{P}^2$ density (d) zoomed in (c)

Fig. 10 The results of the Shu-Osher problem with the OFDG scheme at $t = 1.8$.

$$\rho(r(R,t),t) = \rho_0(R)a(t)^{-\frac{2}{\gamma-1}},$$
$$u^r(r(R,t),t) = R\frac{\mathrm{d}}{\mathrm{d}t}a(t),$$
$$p(r(R,t),t) = p_0(R)a(t)^{-\frac{2\gamma}{\gamma-1}},$$

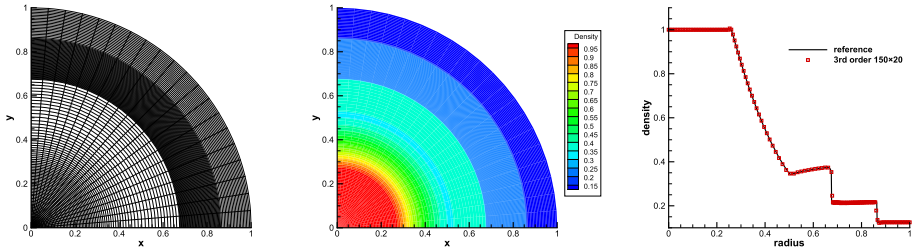
where u^r represents the velocity in the radial direction and $r(R,t)$ represents the radius at time t of a point initially located at radius R. Its analytical solution is $r(R,t) = a(t)R$.

We test the problem on an initially 30×40 grid. The final time is $t = \frac{\sqrt{3}}{2}\tau$ in order to obtain compression rate $a(t) = 1/2$. In Fig. 15, we plot the mesh, density contour and density profile for the Lagrangian type DG scheme. From the figures, we find that the density solution of the third-order scheme is closer to the analytic solution than that of the first-order scheme and the second-order scheme. And the internal and external radii of the third-order scheme are accurately located. We also observe the symmetry-preserving properties of the DG scheme.



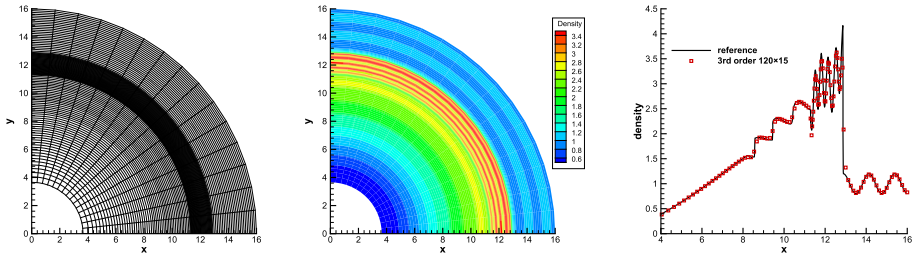
(a) compare (b) zoomed in (a)

Fig. 11 The comparison of results for the Shu-Osher problem with the OFDG scheme.



(a) mesh (b) density (c) density profile

Fig. 12 The results of the polar Sod problem with the $\mathcal{P}^2$ DG scheme at $t = 0.2$.



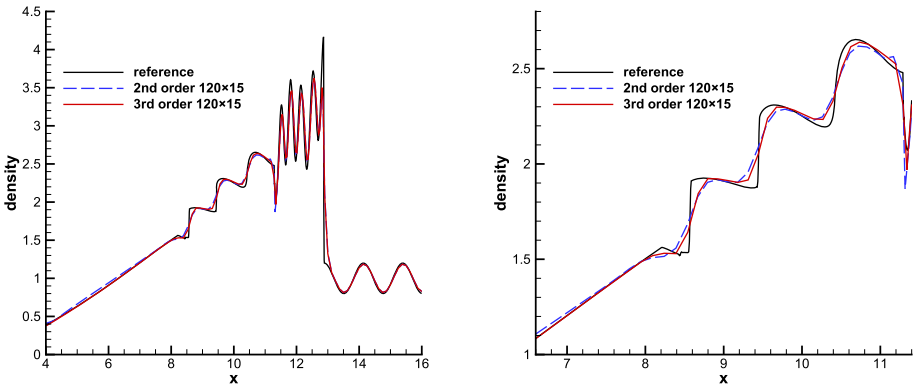
(a) mesh (b) density (c) density profile

Fig. 13 The results of the polar Shu-Osher problem with the $\mathcal{P}^2$ OFDG scheme at $t = 1.8$.

Next, we will test the classic two-dimensional Lagrangian examples, including the Sedov and Saltzman problems.

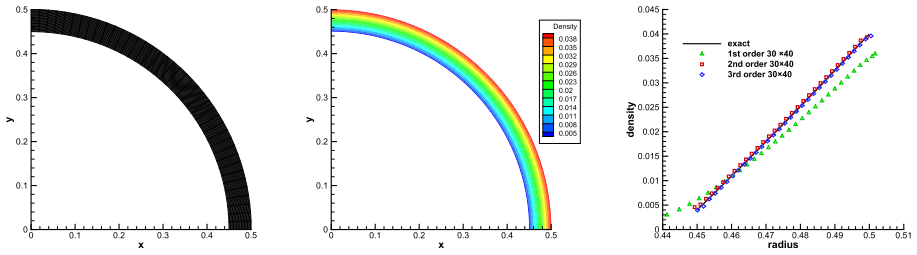
Example 4.10 The Sedov blast wave problem

The Sedov blast wave problem [38] simulates the expanding wave caused by an intense explosion in a ideal gas. The initial domain is defined by $(x, y) \in [0, 1.1]^2$. Consider this problem with initial condition as:



(a) compare (b) zoomed in (a)

Fig. 14 The comparison of results for the polar Shu-Osher problem with the OFDG scheme.



(a) $\mathcal{P}^2$ mesh (b) $\mathcal{P}^2$ density (c) density profile

Fig. 15 The results of the Kidder problem with the DG scheme at $t = \frac{\sqrt{3}}{2}\tau$.

$$\rho_0 = 1,\ u_0 = v_0 = 0.$$

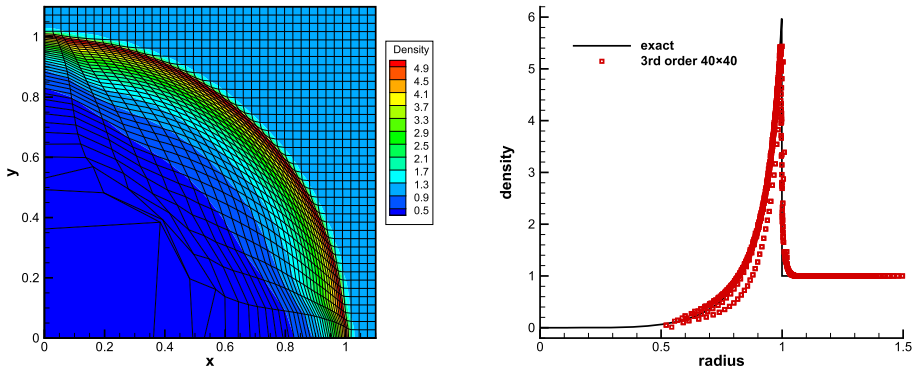
The initial internal energy $e_0 = 10^{-6}$ almost everywhere except for the only one cell I near the origin where we set $e_0 = \frac{0.244816}{|I|}$. We take $\gamma = 1.4$. Reflective boundary conditions are used on the four boundaries. The final time is $t = 1$. The exact solution gives a shock at unit radius with a peak density of 6.

We test this problem on the initially rectangular grid consisting of 40×40 cells. In this case, the shock direction is not aligned with the grid lines. Fig. 16 shows the third-order results at $t = 1$, including the mesh, density contour, and density as a function of the radial radius at all grid points. We can observe that the density solution is very close to the one-dimensional exact solution, and the shock front is correctly located and is almost circular. We can also see that there is no oscillation near the shock. These results indicate that our designed OFDG scheme has good robustness and accuracy.

Example 4.11 The Saltzman problem

The Saltzman problem [14] is a challenging example to evaluate the robustness of Lagrangian schemes when the mesh is not aligned with the fluid flow. The problem consists of a rectangular box whose left side is a piston. The initial condition is set to

$$\rho_0 = 1,\ u_0 = 0, v_0 = 0,\ p_0 = 10^{-4}.$$



(a) density (b) density profile

Fig. 16 The results of the Sedov problem with the $\mathcal{P}^2$ OFDG scheme at $t = 1$.

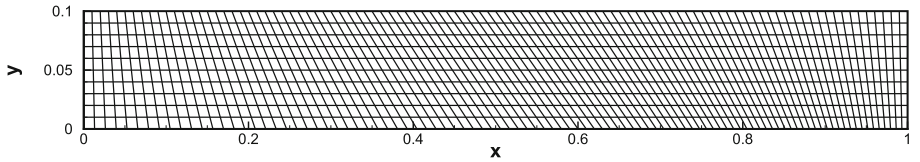


Fig. 17 The initial mesh of the Saltzman problem.

The gas constant is $\gamma = 5/3$. The left boundary of the computation domain is a moving piston with a constant velocity of 1. Reflective boundary conditions are used on the right, upper and lower boundaries. The final time is $t = 0.6$. The analytical post shock density is 4.0 and the shock speed is $\frac{4}{3}$. At $t = 0.6$, the analytical shock is located at $x = 0.8$.

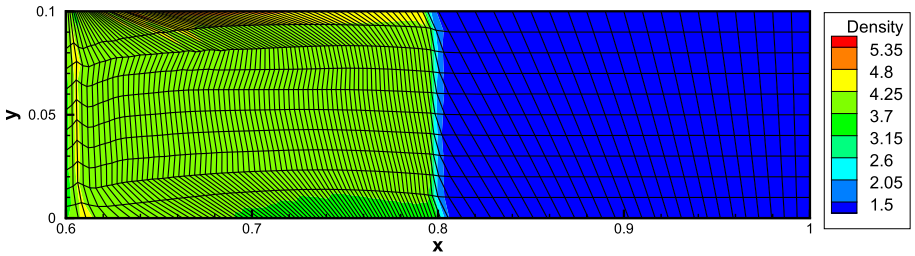
The initial grid is displayed in Fig. 17, defined by 100 cells in the x-direction and 10 cells in the y-direction, with the following expression

$$x(i, j) = (i - 1)\Delta x + (11 - j)\sin(0.01\pi(i - 1))\Delta y, \quad y(i, j) = (j - 1)\Delta y,$$

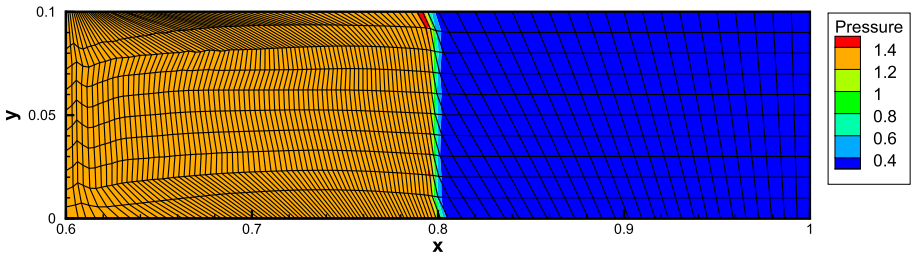
where $\Delta x = \Delta y = 0.01$. For this test, it is necessary to first use a smaller Courant number in order to maintain stability. The Courant number λ is set to be 0.01 initially and returns to be 0.3 after $t = 0.01$. The meshes, density solution and pressure solution at $t = 0.6$ are shown in Fig. 18. It can be observed that our designed Lagrangian type OFDG scheme can effectively maintain one-dimensional solutions except for the results near the upper and lower boundaries being affected by boundary conditions. The shock position is in good agreement with the analytical position.

5 Concluding remarks

In this paper, we have constructed a high-order conservative Lagrangian type DG scheme coupled with the SSP-RK time discretization for solving the compressible Euler equations on the straight-edge quadrilateral meshes. This scheme belongs to the cell-centered updated Lagrangian formulation, where the DG space is defined in the physical coordinates, and the conserved variables are solved directly. Compared to previous research, our proposed



(a) density



(b) pressure

Fig. 18 The results of the Saltzman problem with the $\mathcal{P}^2$ OFDG scheme at $t = 0.6$.

Lagrangian type DG scheme can achieve arbitrarily high order accuracy for the smooth solutions on the straight-edge meshes instead of the curved meshes, thus it is simple to implement and can save computational cost effectively. Our scheme also avoids the difficulty of mesh rezoning and physical variable remapping on the curved meshes. Moreover, we have designed the $\mathbf{u} - \mathbf{u}_g$ flux term of this scheme according to the strategy in our previous second-order pure Lagrangian scheme at the material interfaces, contact discontinuities and shocks, so that they can be captured automatically and sharply. To suppress spurious oscillations, we have introduced the damping equation into the Lagrangian type DG method. The damping term is designed to be suitable for moving non-uniform grids in the Lagrangian type framework. A variety of numerical experiments demonstrate that our high-order Lagrangian type DG schemes have the optimal order accuracy, can effectively control the numerical oscillation, and maintain good resolution at the discontinuities. Although we have only considered quadrilateral meshes in this paper, the extension to arbitrary unstructured meshes is straightforward. In the future, we would like to further combine appropriate mesh rezoning step and physical variable remapping step on the straight-edge meshes to avoid large mesh deformations in numerical simulations of more complex and realistic physical models.

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